RAPTOR: Routing Attacks on Privacy in Tor

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Course: Seminar Cyber

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Can Tor users stay anonymous when the network is watching?



Tor aims at preventing adversaries to follow packets between a sender and a receiver



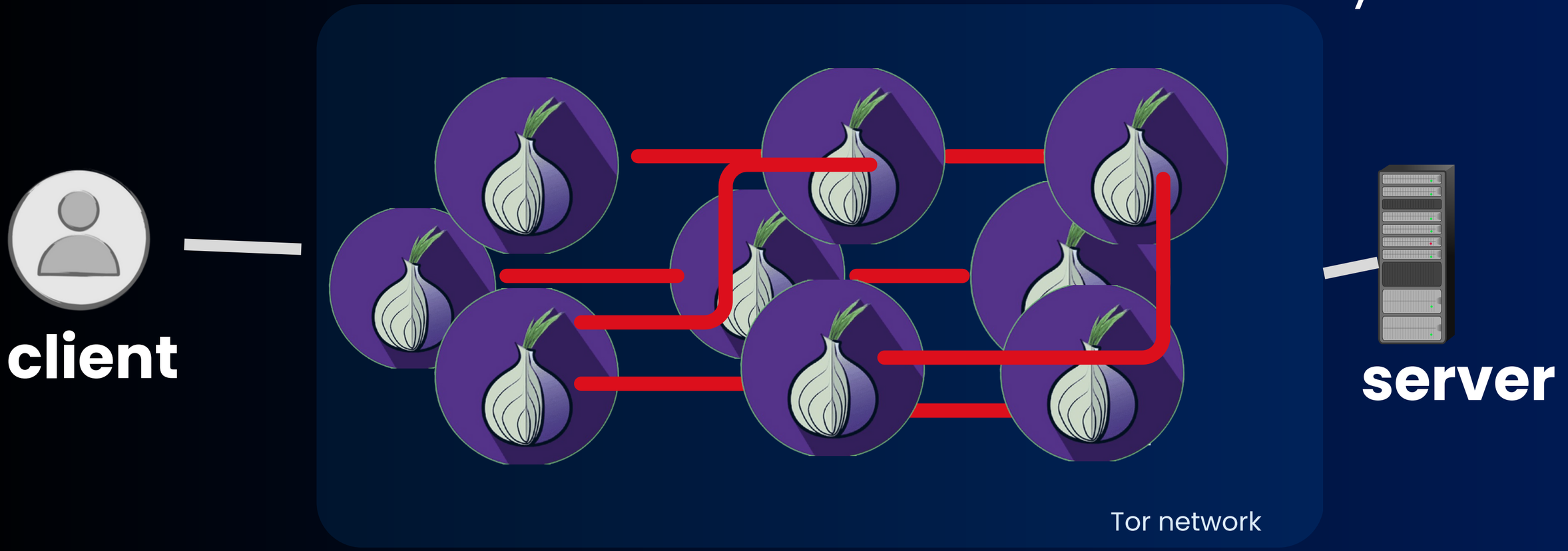
client

Tor network

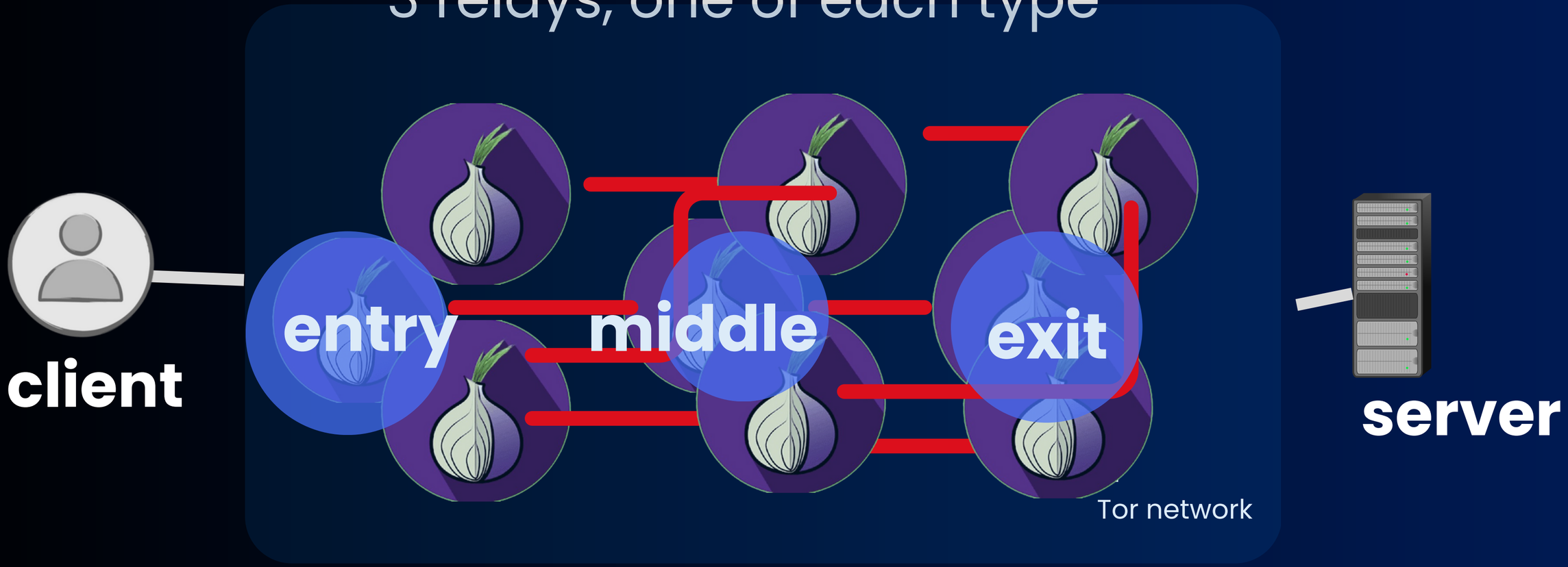


server

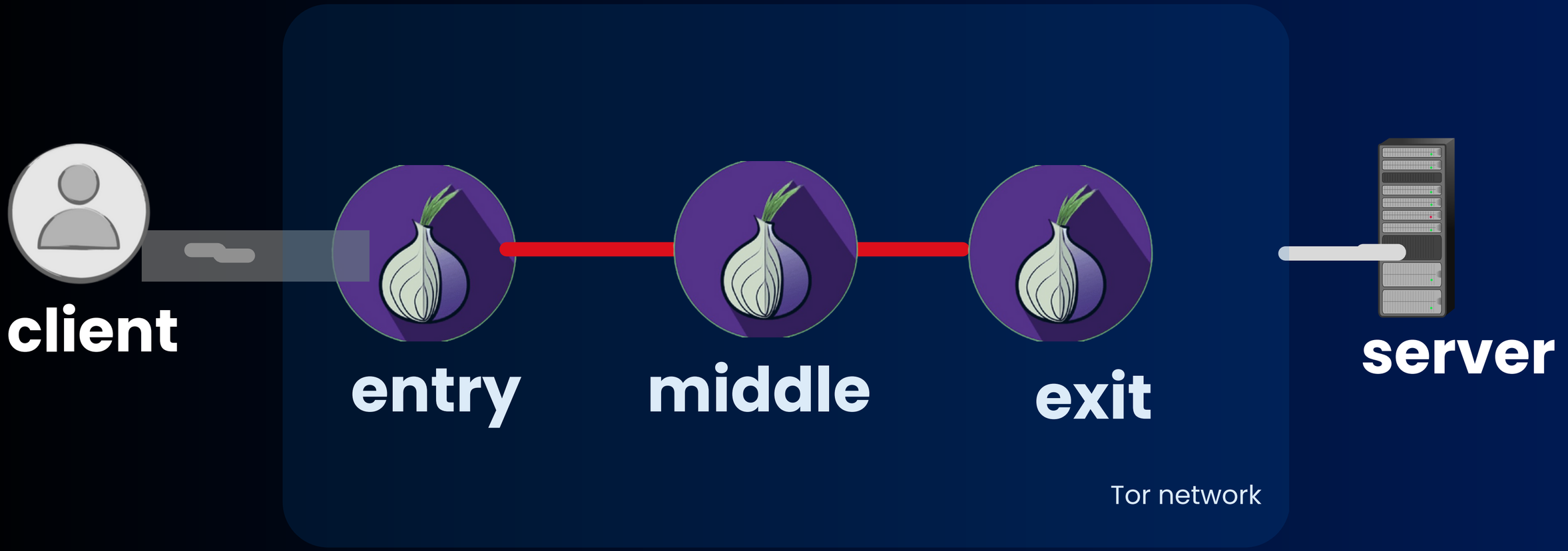
To do that,
Tor bounces traffic around a network of relays



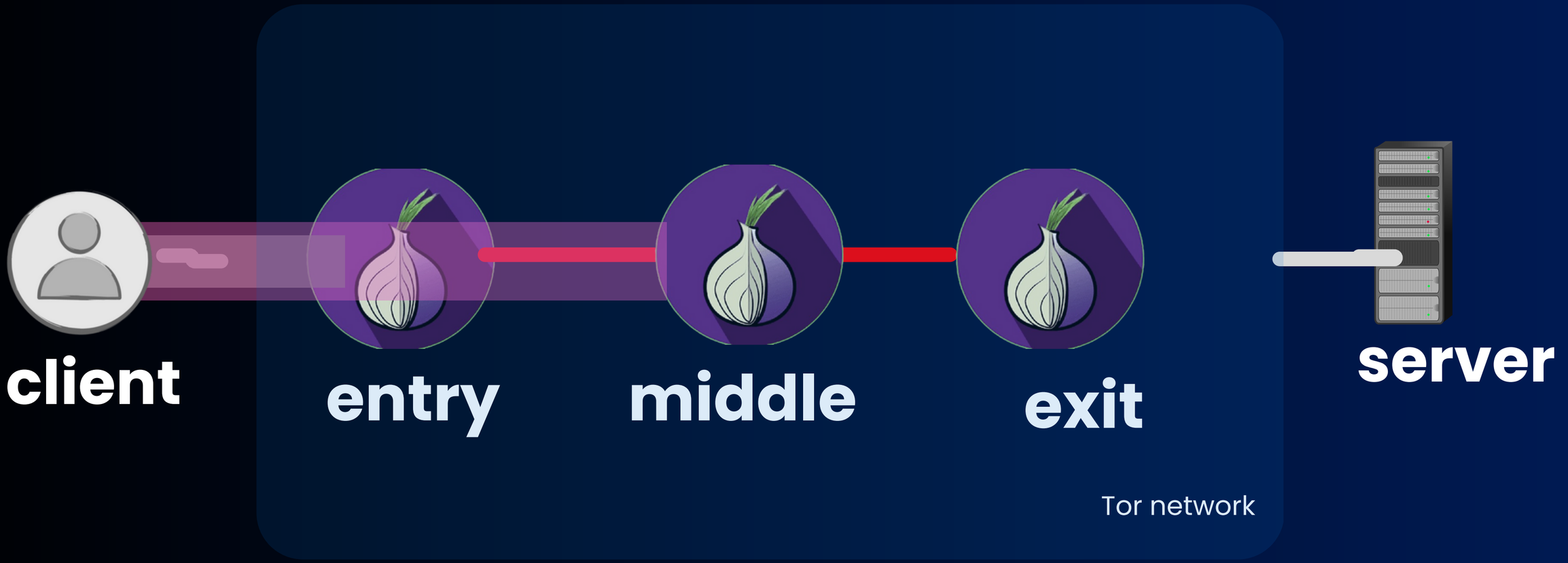
Tor clients start by selecting
3 relays, one of each type



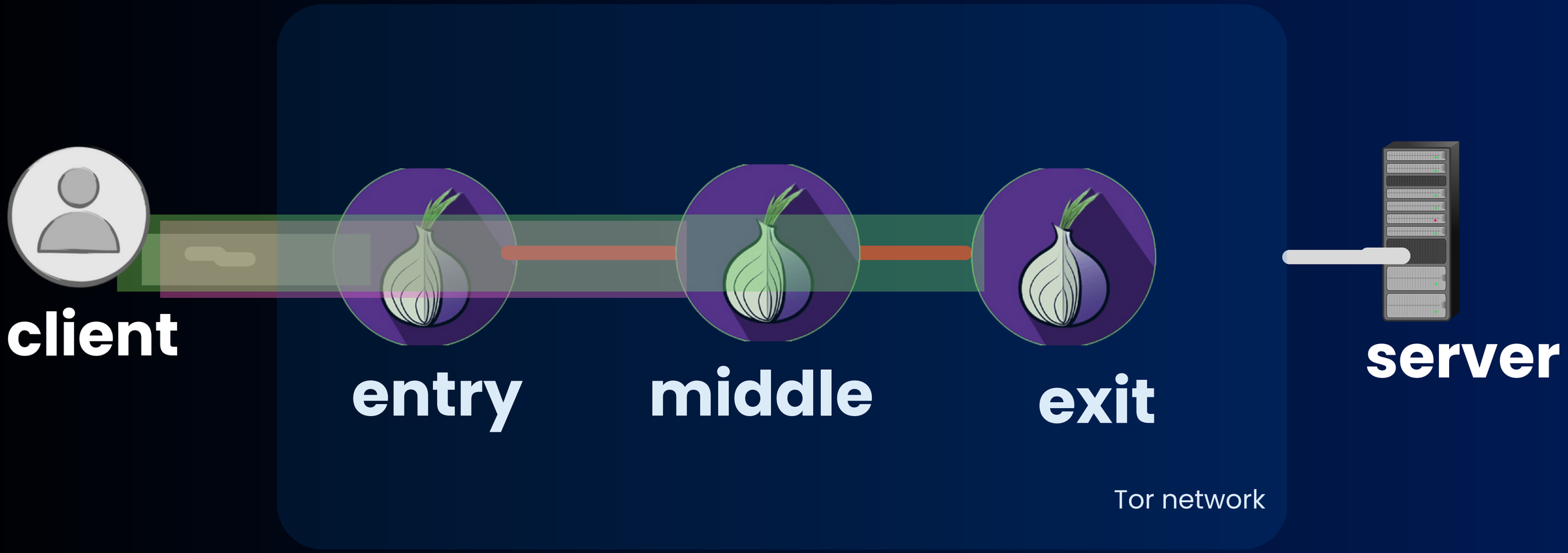
Tor clients then incrementally build encrypted circuits through them



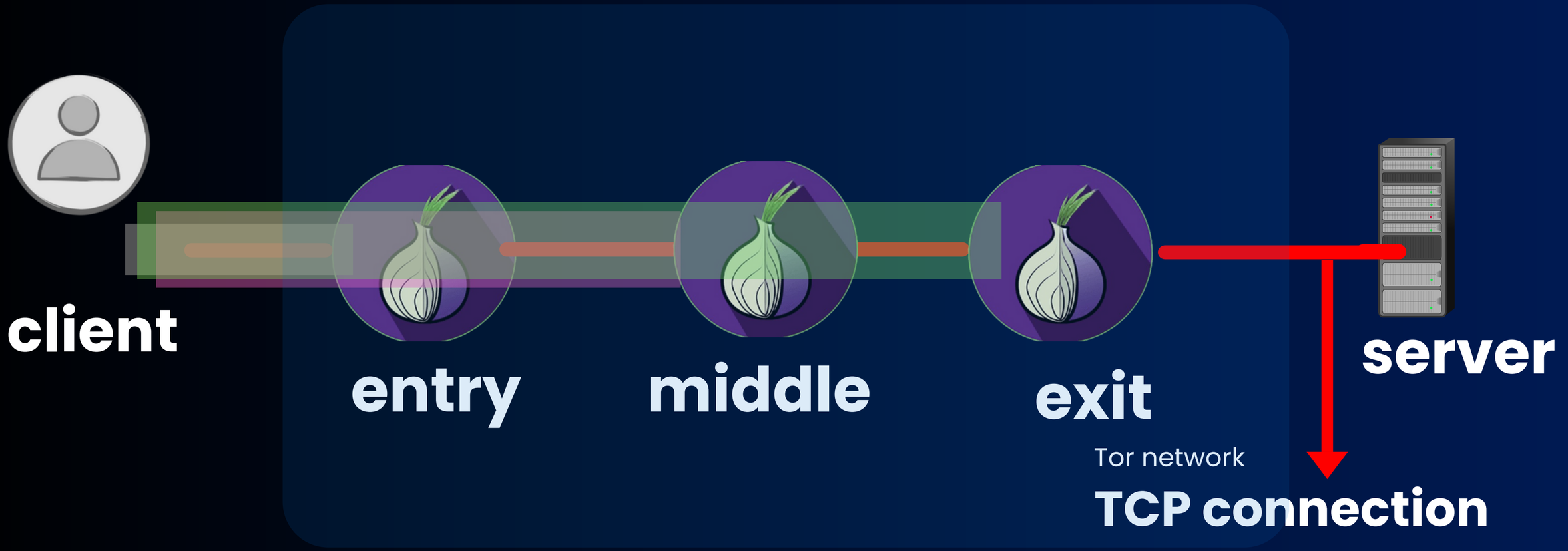
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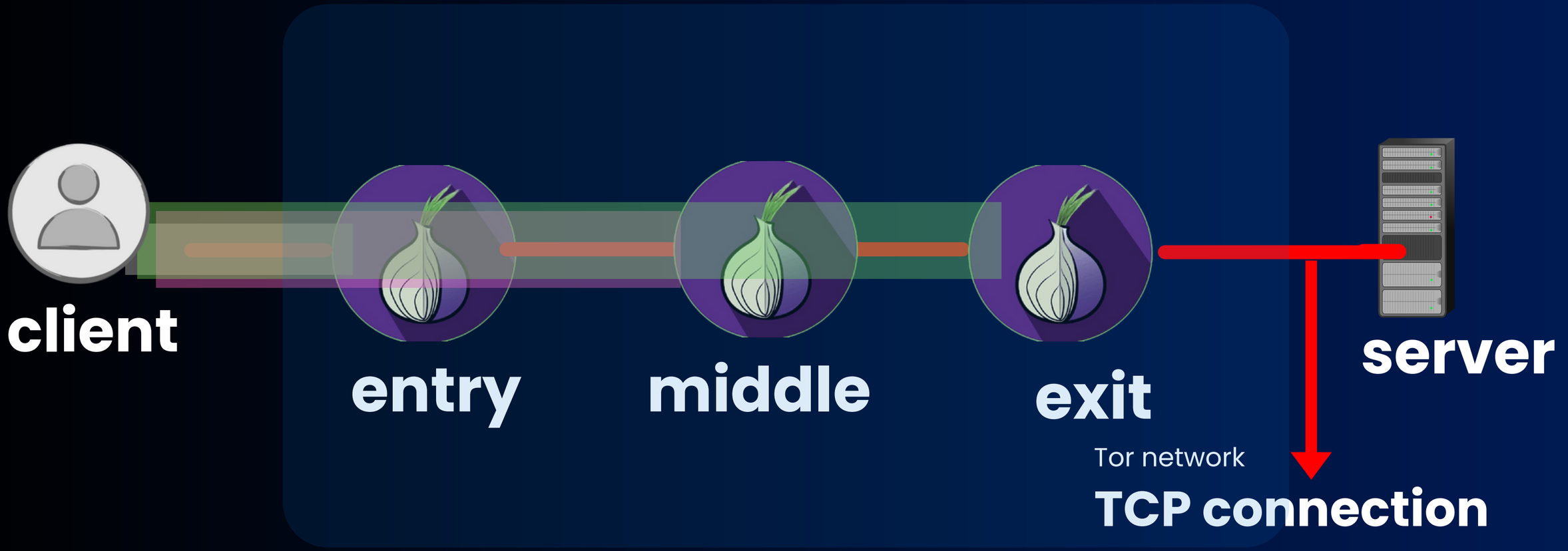
Tor clients then incrementally build encrypted circuits through them



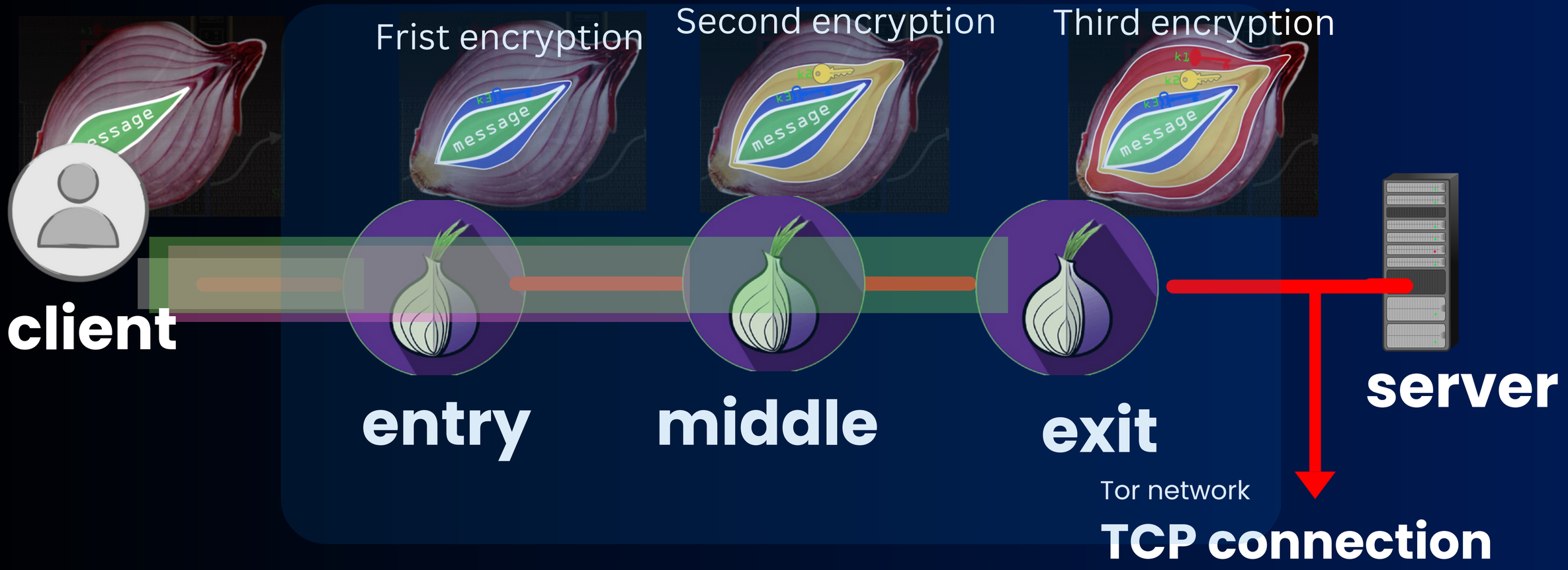
Tor clients then incrementally build encrypted circuits through them



Anonymous communication takes place by forwarding across consecutive tunnels



Not a single Tor entity knows both client and server





People trust Tor because it's open-source, decentralized, and uses layered encryption.

But does this guarantee privacy against all types of threats?

Threat Model

Tor assumes that no single entity can observe both ends of the communication.

But what if the network infrastructure itself is under attack?

thers two ways



Is This Really a Threat?

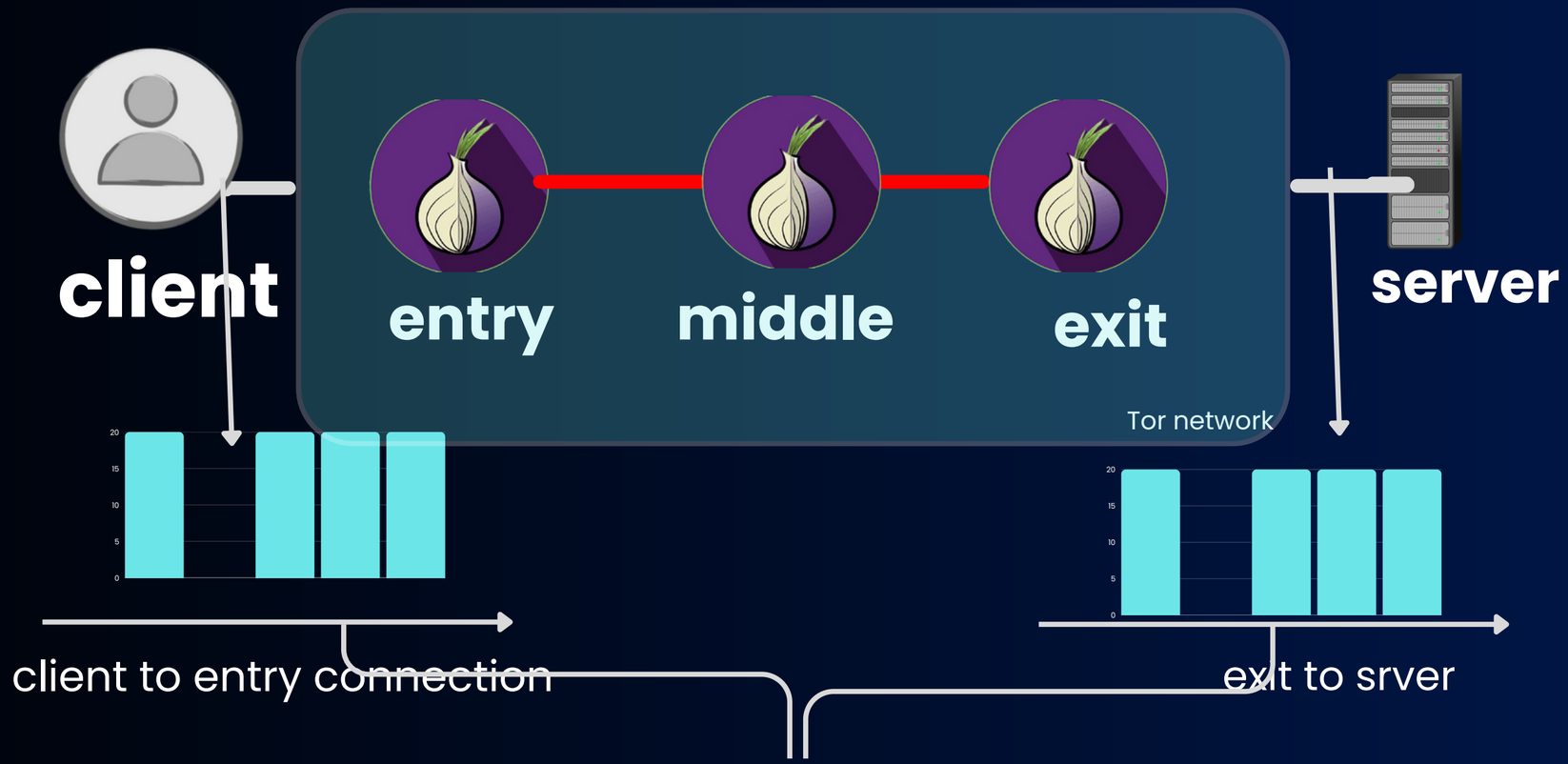
- The internet is routed using **BGP** (Border Gateway Protocol).
- BGP is **dynamic, often asymmetr**ic, and has no built-in security.
- In fact, **many ASes already observe Tor traffic — by design or by chance.**

Tor encrypts the content… but not the patterns.

Even without breaking encryption, traffic timing and volume can still leak metadata.

That’s how AS-level adversaries perform traffic correlation.

Correlation Attacks

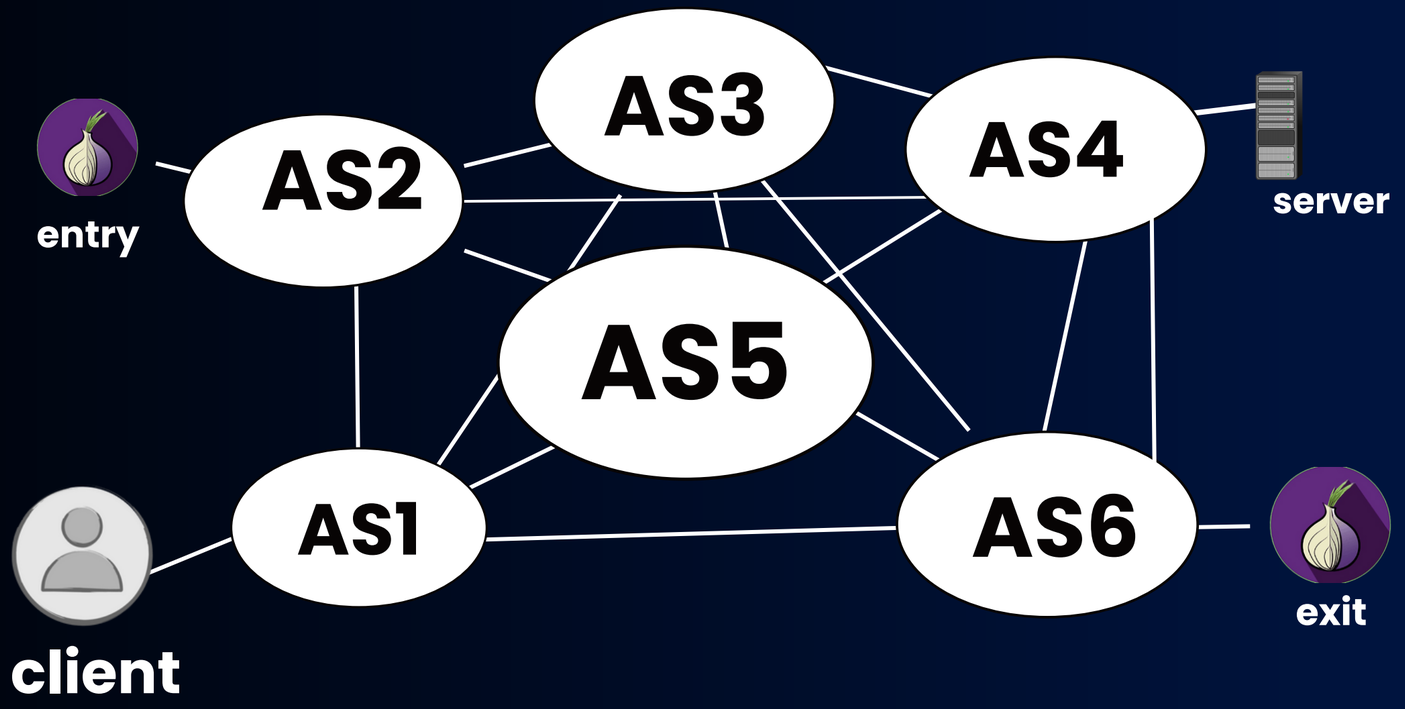


Traffic correlation attacks require to see client-to-entry and exit-to-server traffic

How?

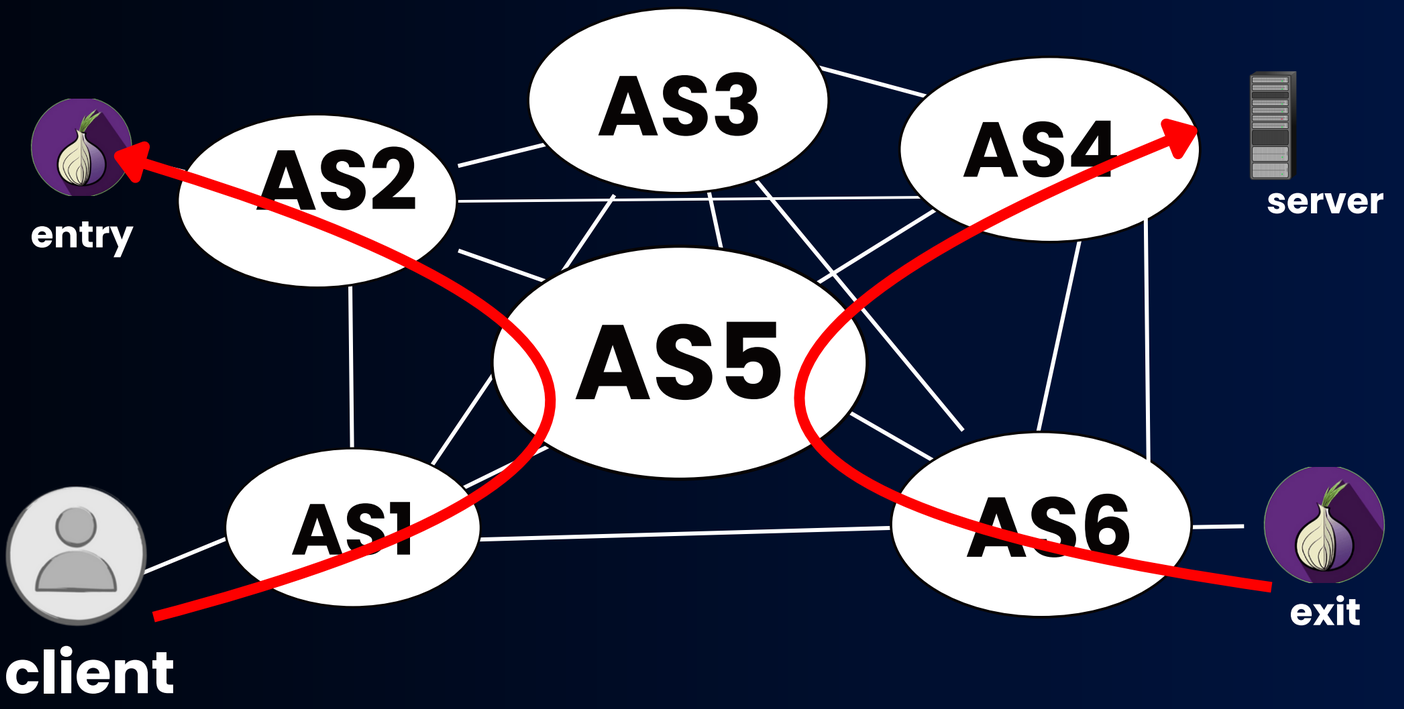
Correlation Attacks

To break anonymity, attackers don’t need to decrypt anything.



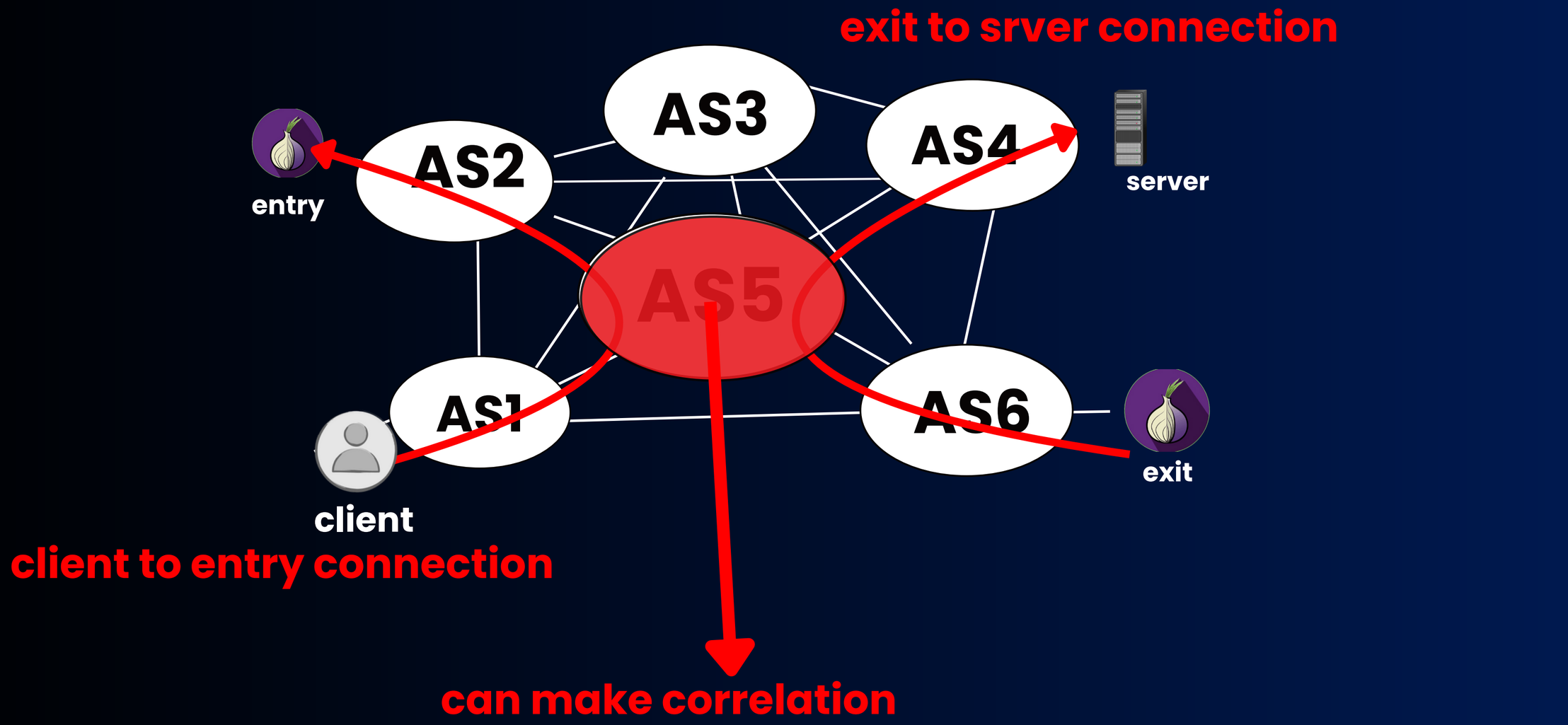
They only need to observe traffic timing on both ends

exit to srver connection



client to entry connection

Tor connections get routed according to BGP



RAPTOR Attacks

User anonymity decreases over time due to BGP dynamics

- **Asymmetric routing**

Traffic from A → B may take a different path than B → A

- **Natural BGP convergence**

Changes in routing caused by policy updates, network failures, or maintenance.

- **Active BGP manipulation**

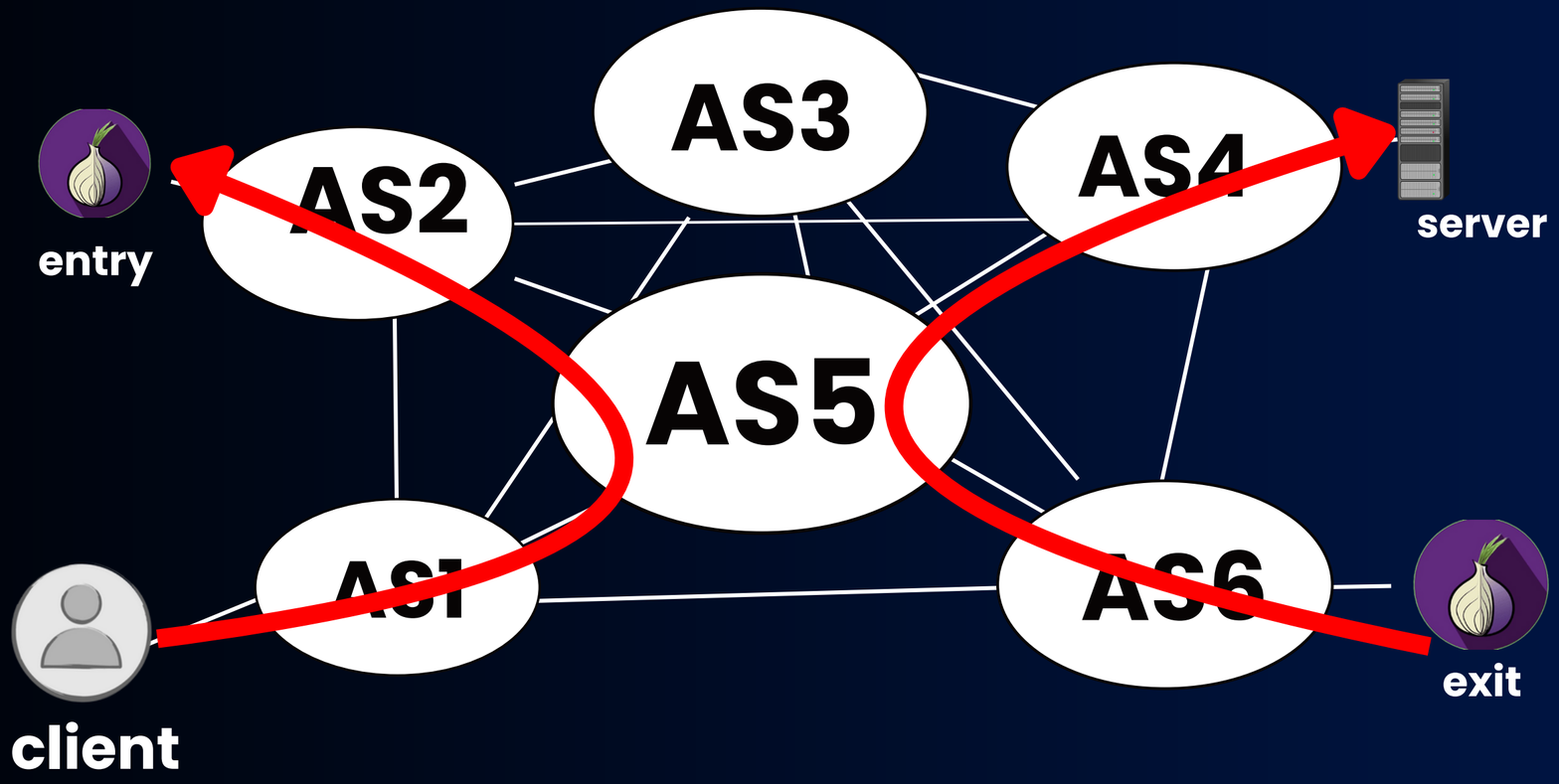
Malicious ASes can hijack or intercept IP prefixes
This enables man-in-the-middle (MITM) attacks on live Tor traffic.

RAPTOR Attack #1

Asymmetric routing

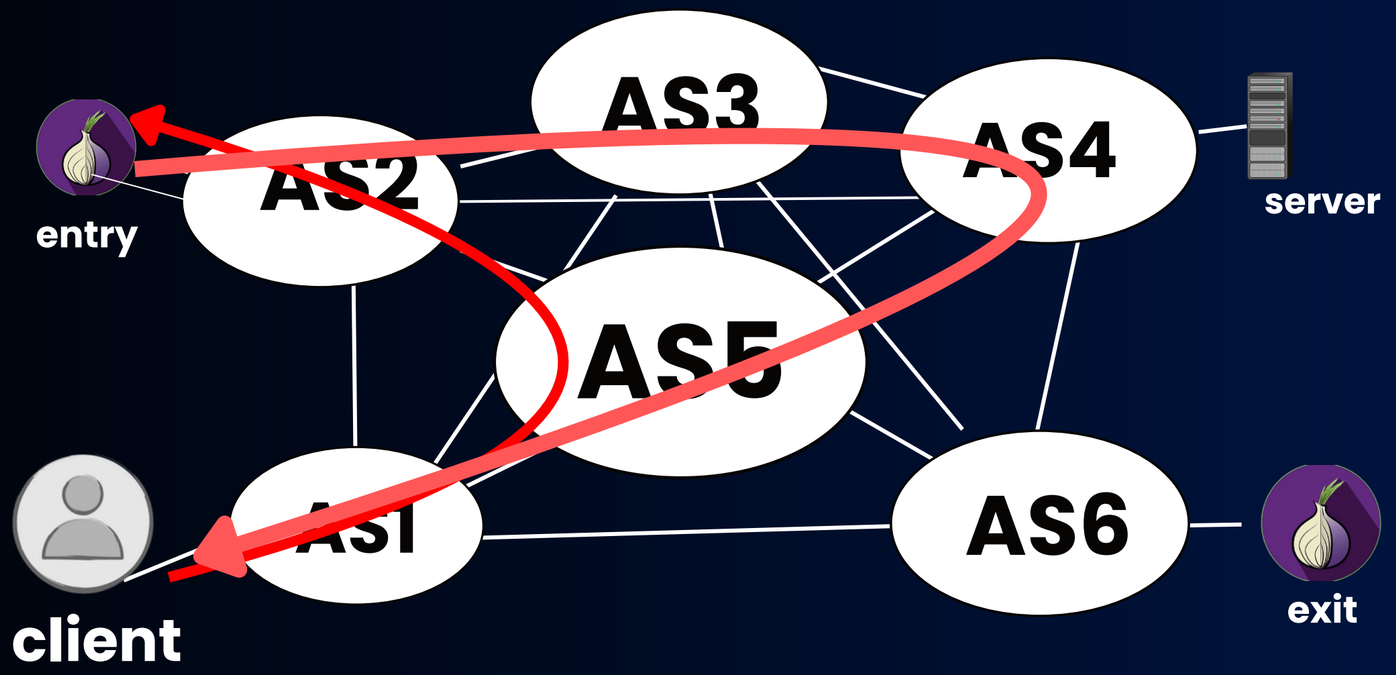
RAPTOR Attack #1

So far, we have considered one side of Tor traffic: client-to-entry and exit-to-server server



RAPTOR Attack #1

However, because of policies, routing is often asymmetric

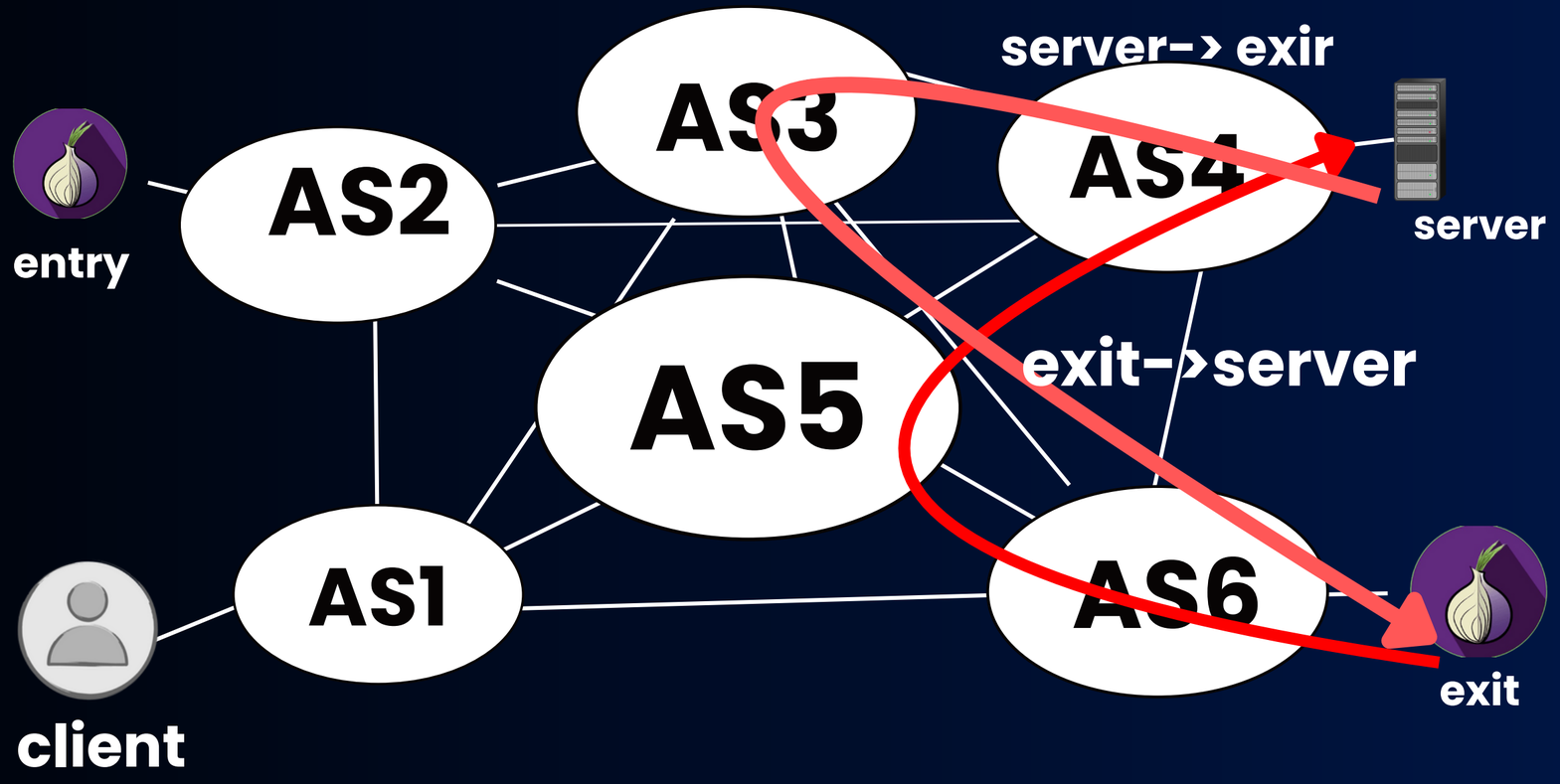


While AS4 does not see client-to-entry traffic,

AS4 sees entry-to-client traffic

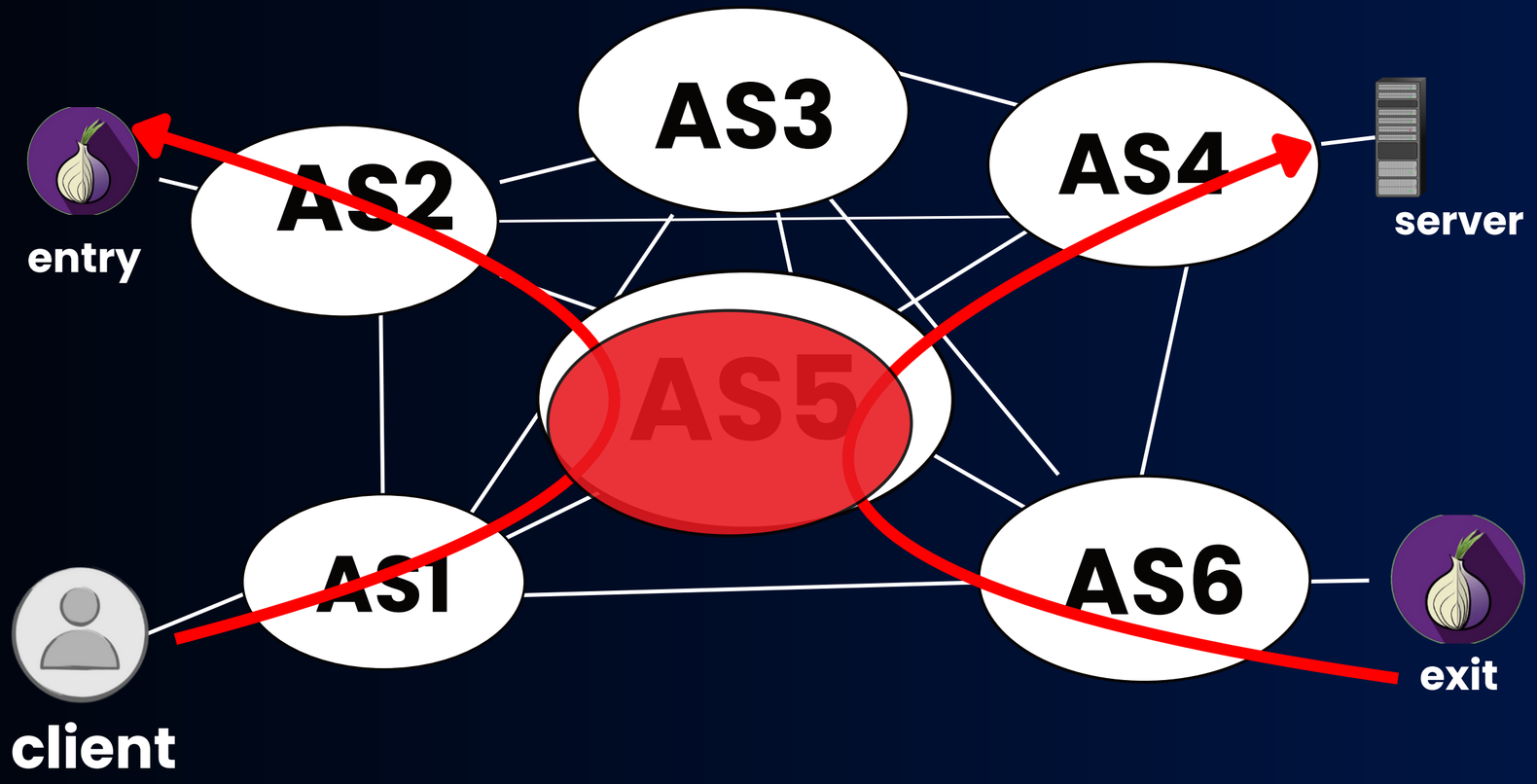
RAPTOR Attack #1

also in server-to-exit traffic



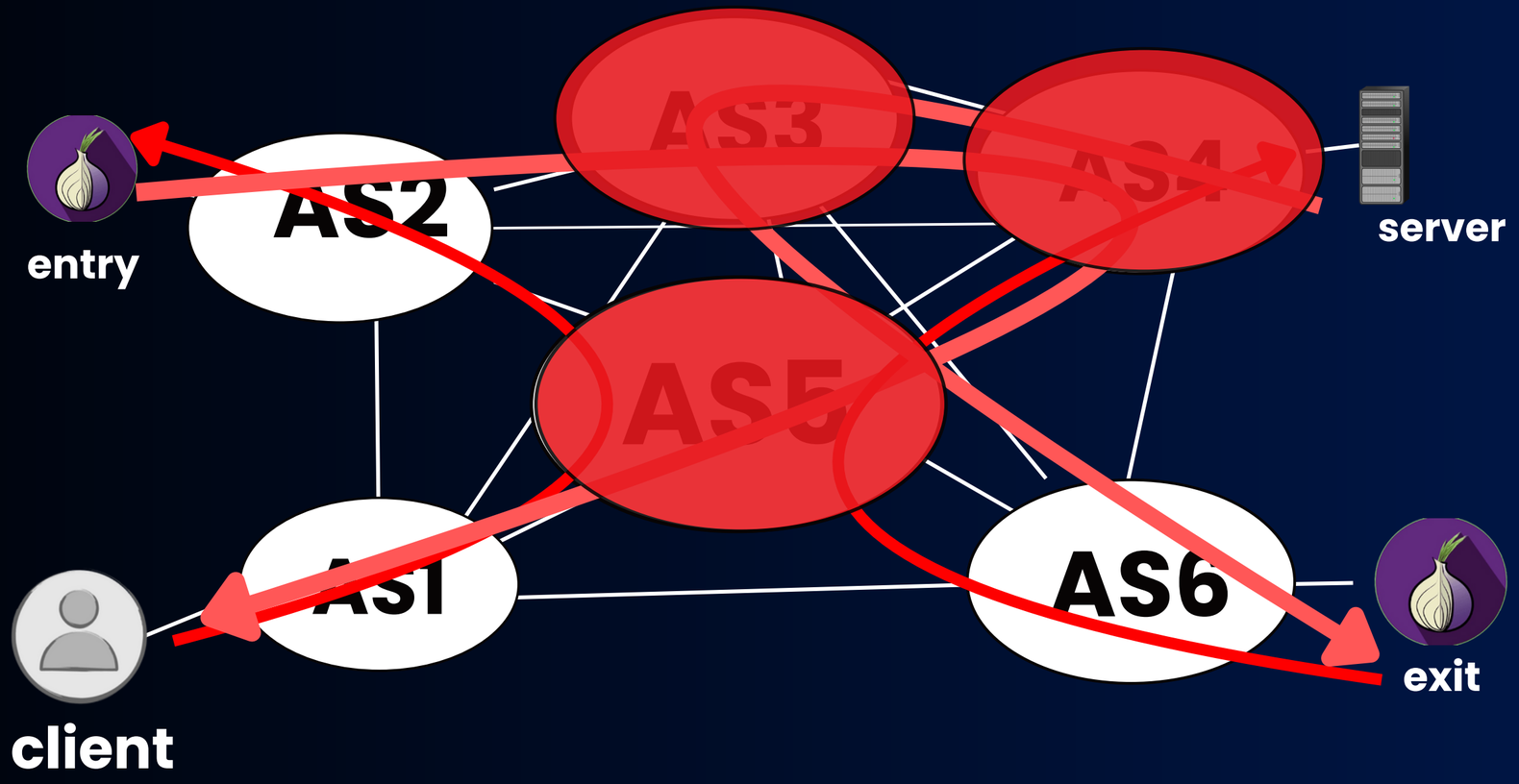
RAPTOR Attack #1

Considering only one direction,
only AS5 is potentially compromising



RAPTOR Attack #1

looking at both directions,
AS3, AS4 and AS5 are potentially compromising



RAPTOR Attack #1

Even if an attacker sees traffic in only one direction, it's almost as useful as seeing both directions.

(e.g., data packets)
Seq: 8282, ACK: 392

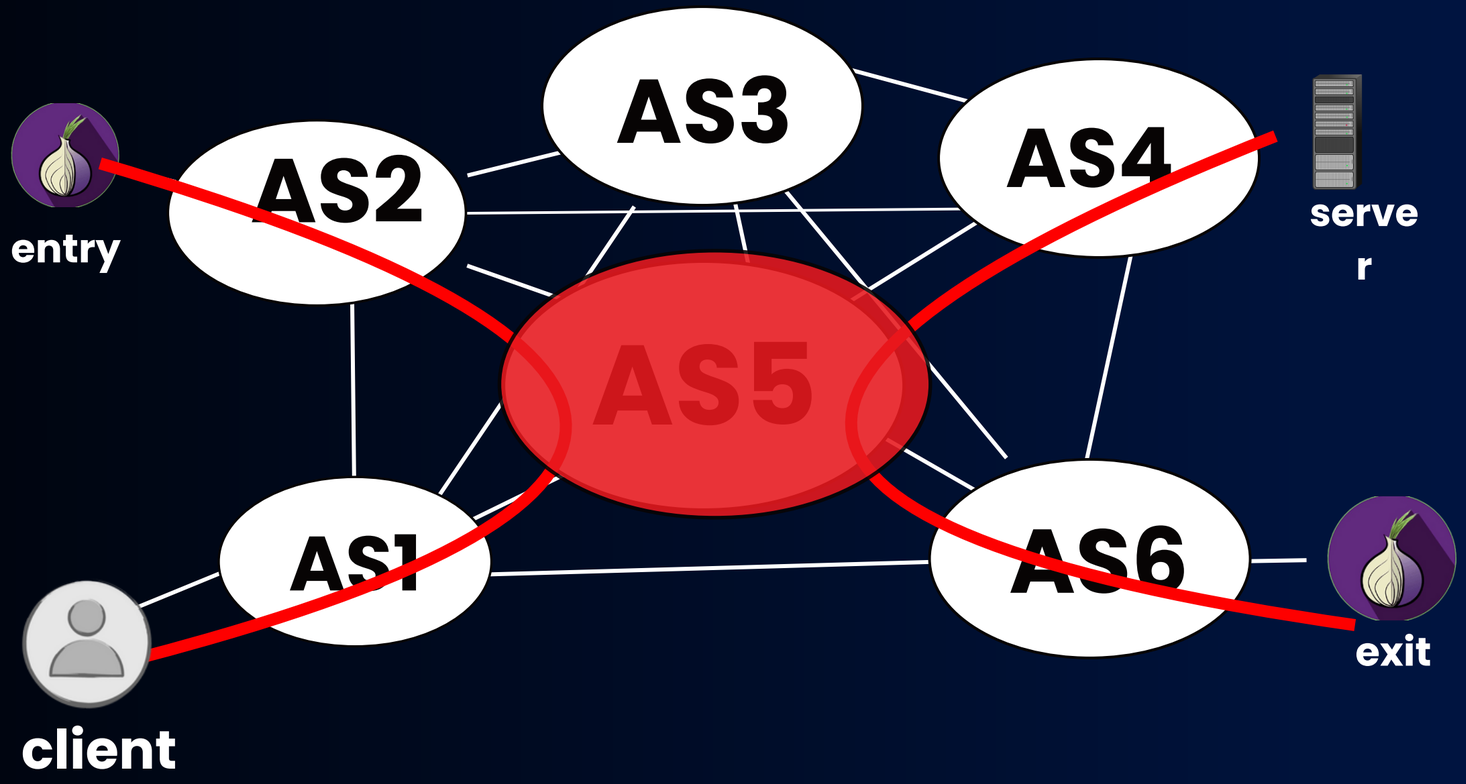
(ACKs & data packets)
Seq: 392, ACK: 8282

RAPTOR Attack #2

Natural BGP churn

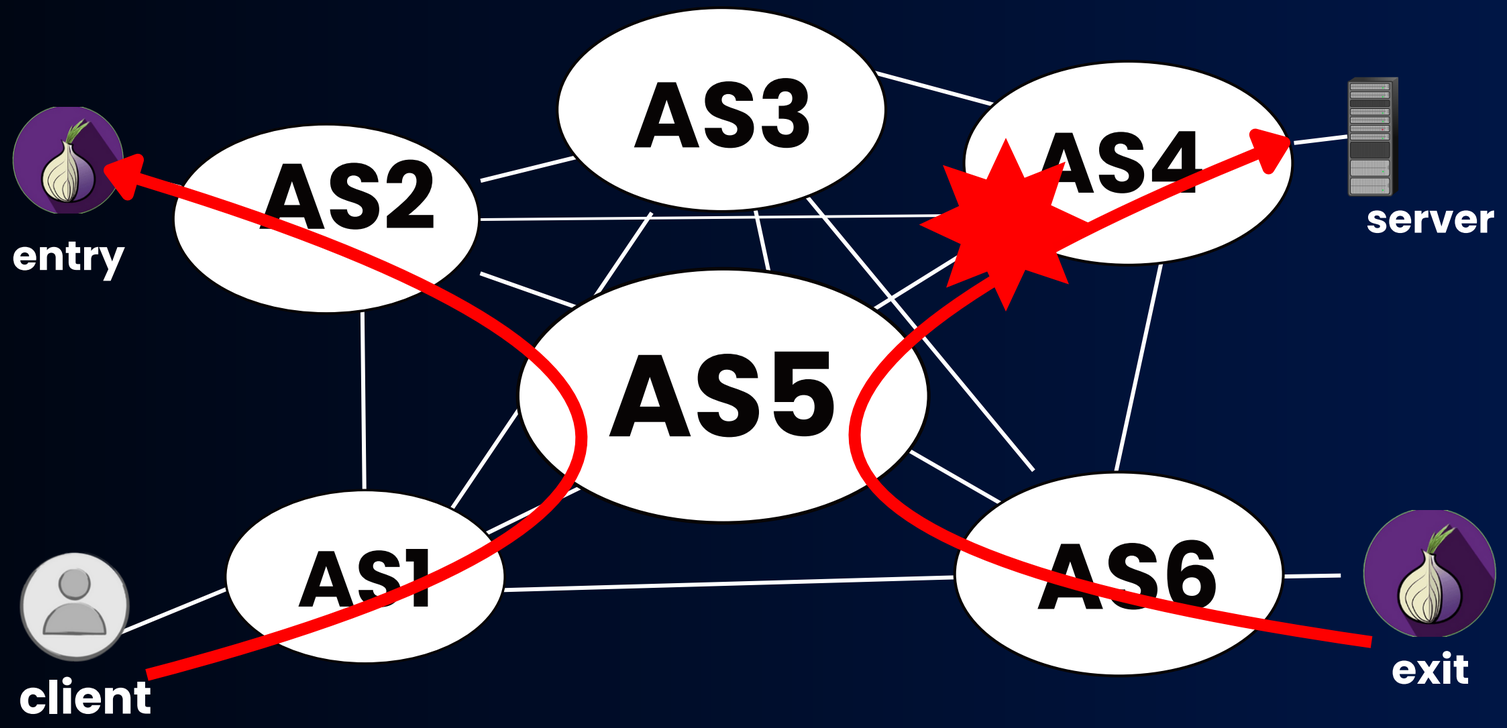
RAPTOR Attack #2

Initially, only AS5 is compromising



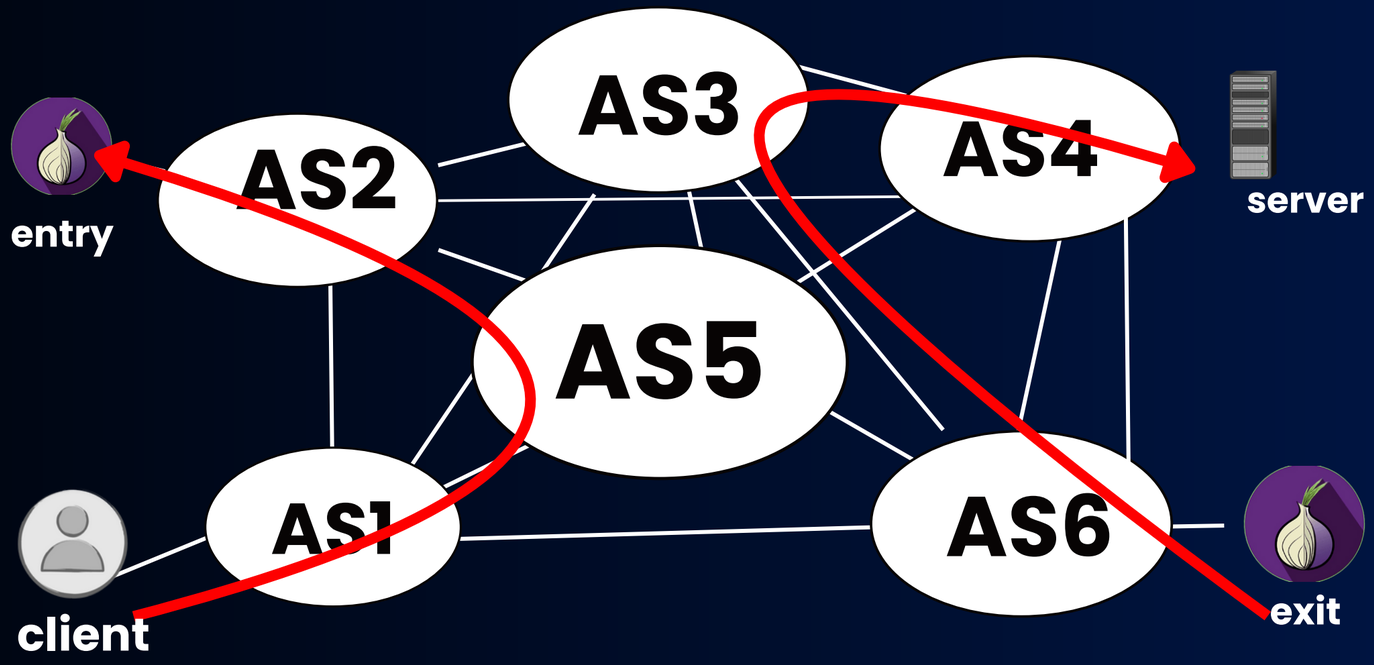
RAPTOR Attack #2

Assume that the link between AS4 and AS5 fails



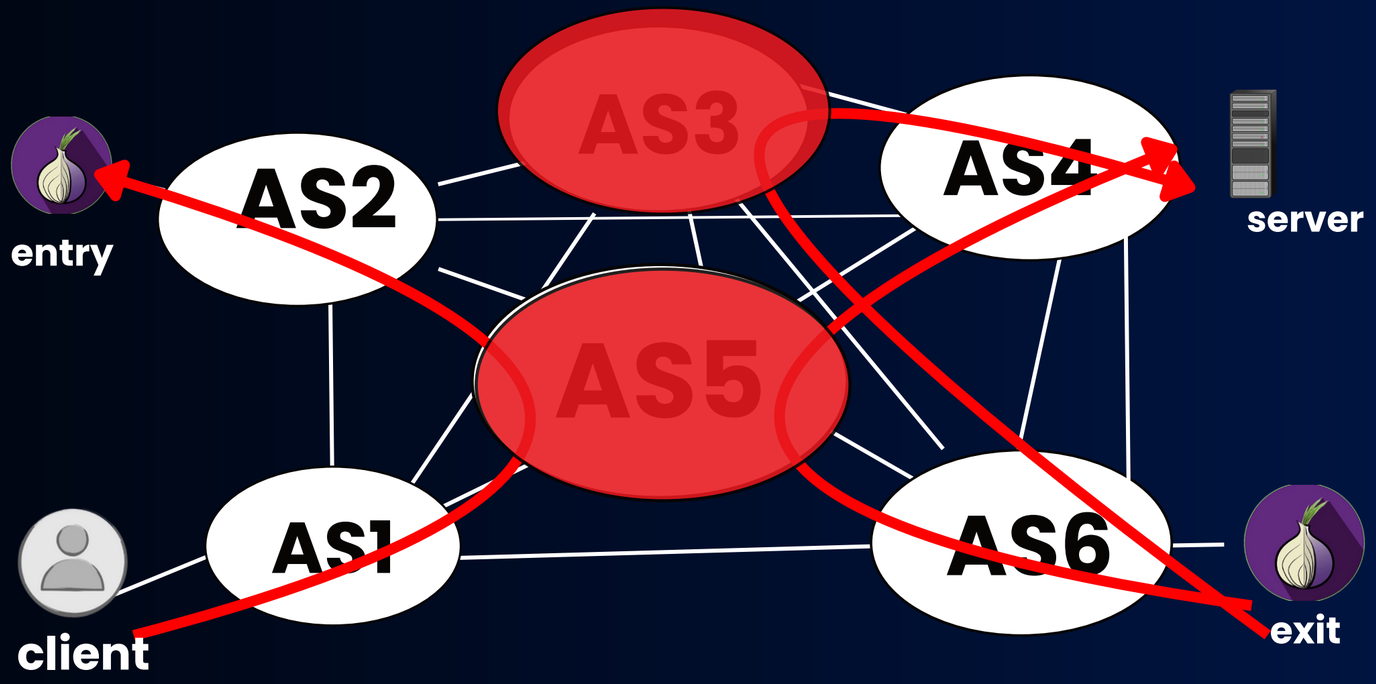
RAPTOR Attack #2

Traffic gets rerouted via AS3



RAPTOR Attack #2

Now, both AS3 and AS5 are seeing client-to-entry and exit-to-server traffic

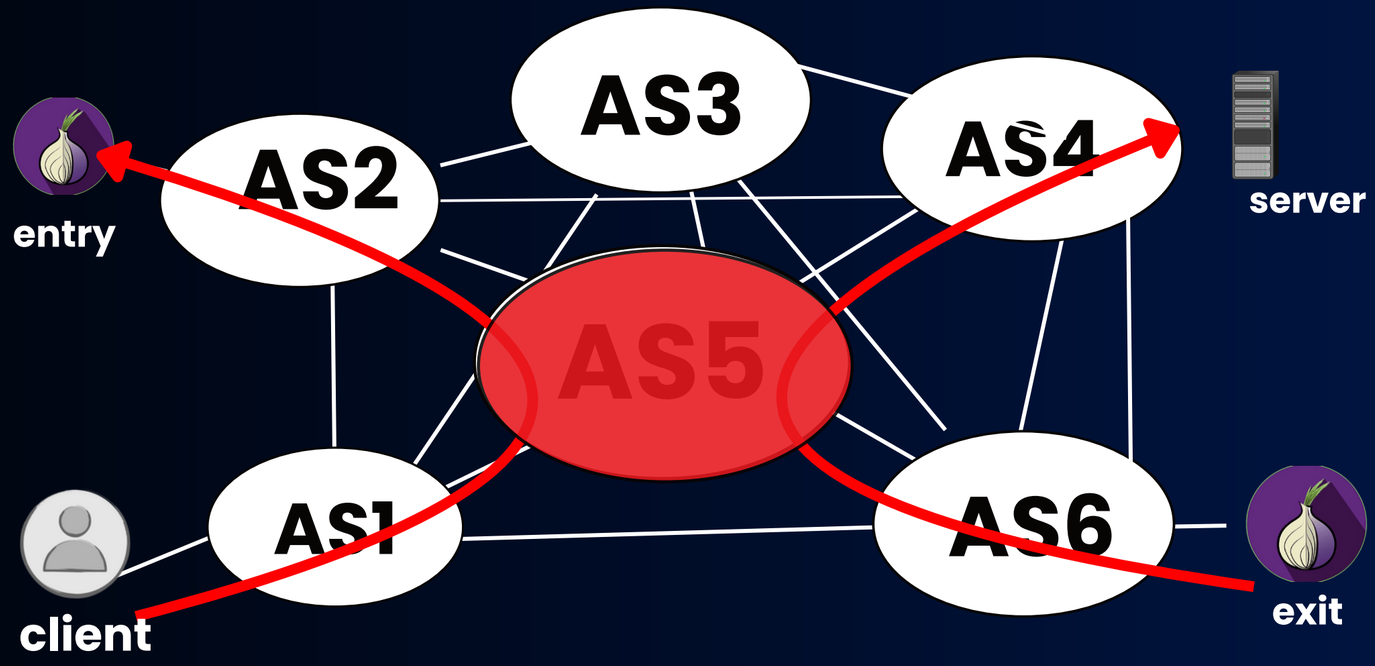


RAPTOR Attack #3

Active BGP manipulation

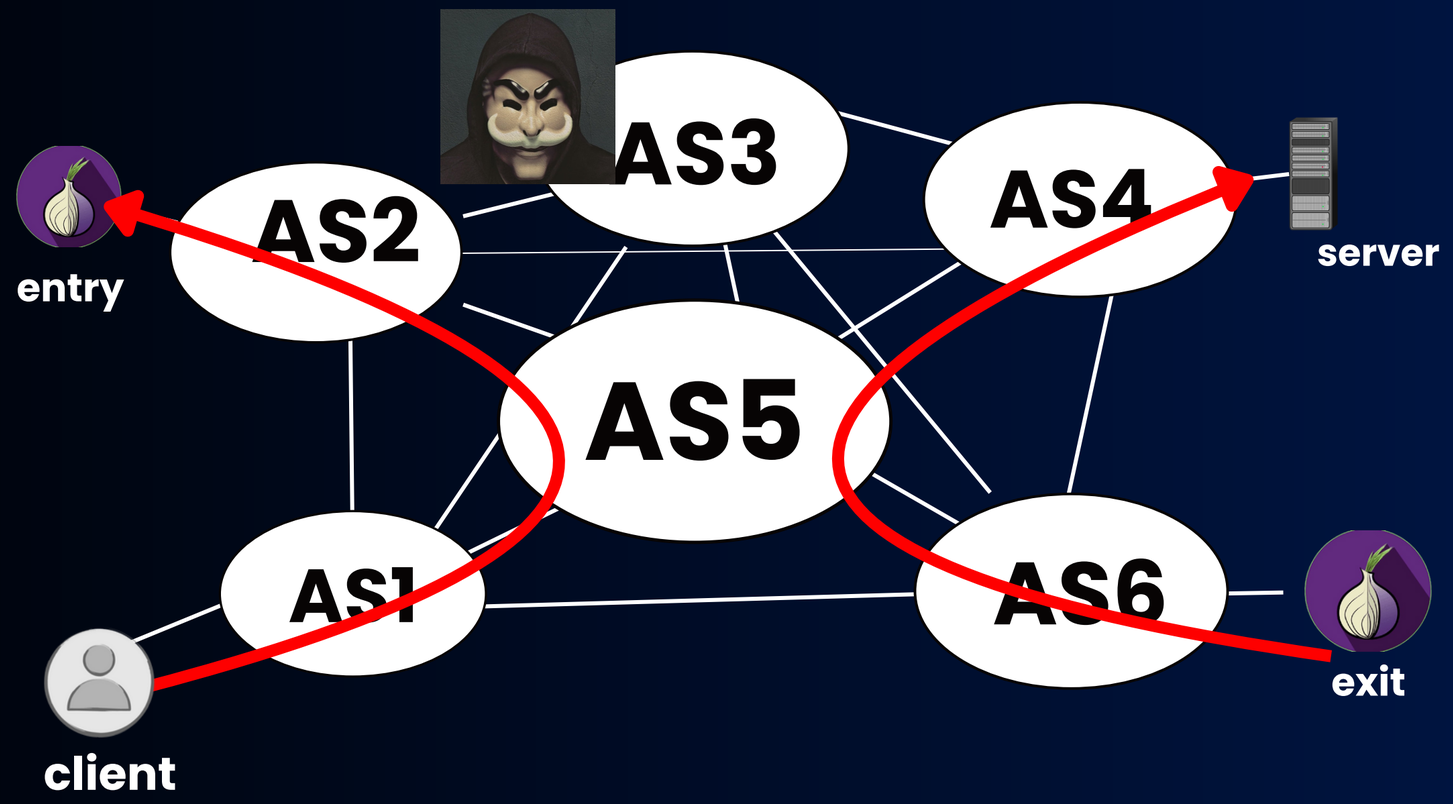
RAPTOR Attack #3

Initially, only AS5 is compromising



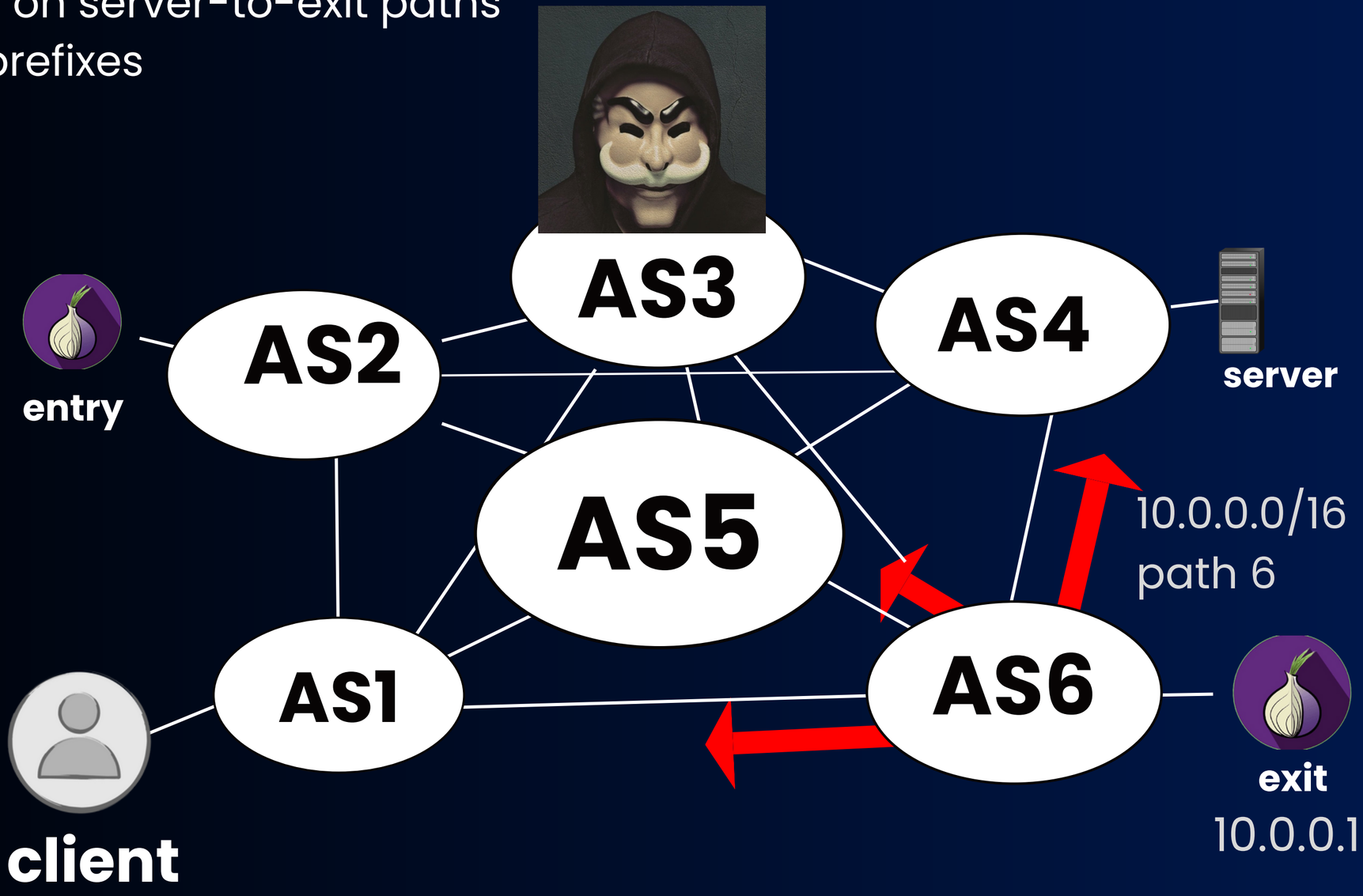
RAPTOR Attack #3

Assume that AS3 is a malicious AS, and wants to observe Tor traffic



RAPTOR Attack #3

AS3 can put itself on server-to-exit paths
by hijacking Tor prefixes



RAPTOR Attack #3



10.0.0.0/24
path 3 2 5 6

AS3

AS2

AS4

AS5

AS1

AS6

entry

server

10.0.0.0/16
path 6

exit

10.0.0.1

client

Testing the Theory – RAPTOR Experiments

We begin with two passive experiments that examine how Tor anonymity is affected by dynamic behavior in the Internet — without any active attacks.

Experiment 1 – Asymmetric Traffic: What if an AS sees only one direction of the flow?

Experiment 2 – BGP Churn: Can routing changes over time expose more users to the same AS?

Experiment 3 — TCP Flow Matching After Hijack

RAPTOR Attack #1

Asymmetric routing

Experimental Validation: Asymmetric Analysis Works

RAPTOR Attack #1

Asymmetric routing

Experiment 1 – Asymmetric Traffic



Each client downloaded a **100MB file** via the Tor network

RAPTOR Attack #1

Asymmetric routing

The goal: capture and analyze TCP ACK traffic

Only one direction of the traffic was observed (client to guard OR exit to server)

RAPTOR Attack #1

Asymmetric routing

results:

	client ACK $ server ACK	client ACK $ server data	client data $ server ACK	client data $ server data
detection rate	96%	94%	96%	94%
false negative	4%	6%	4%	6%
false positive	0%	0%	0%	0%

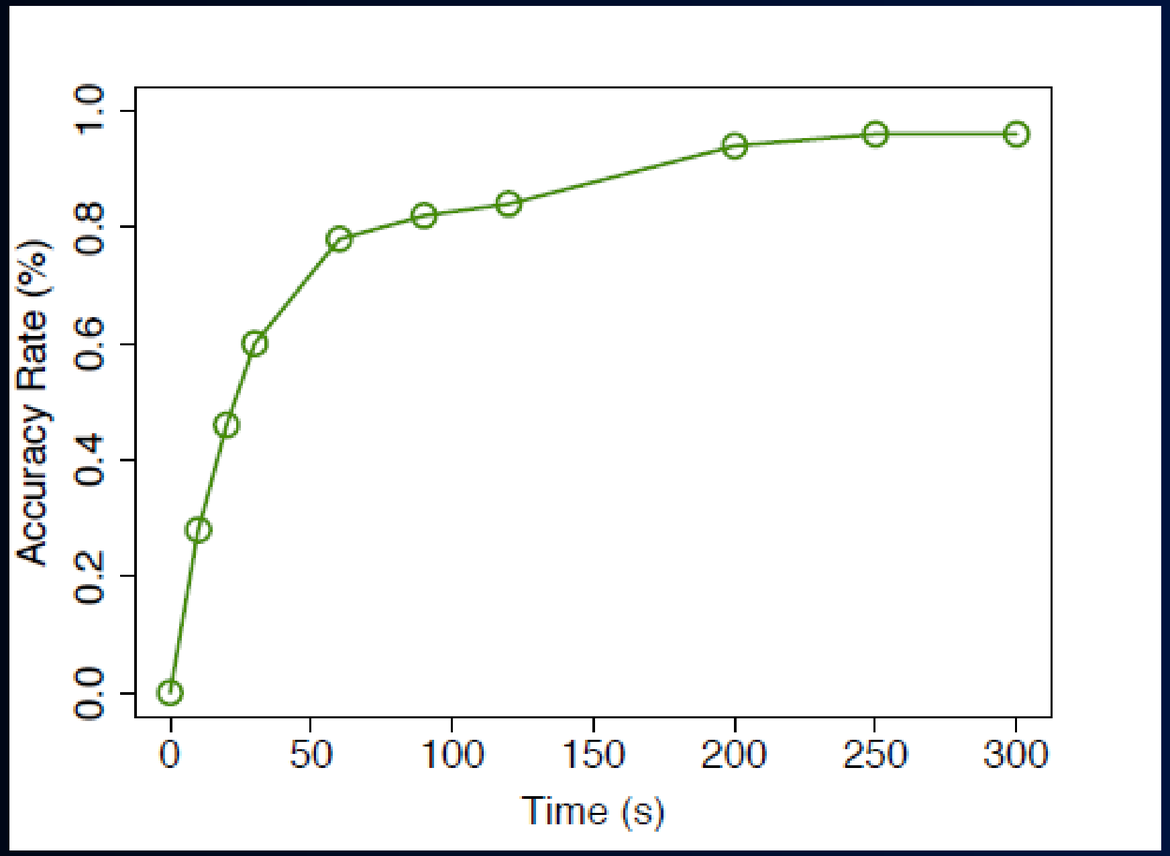
95% of connections were deanonymized in under 5 minutes

- No false positives

RAPTOR Attack #1

Asymmetric routing

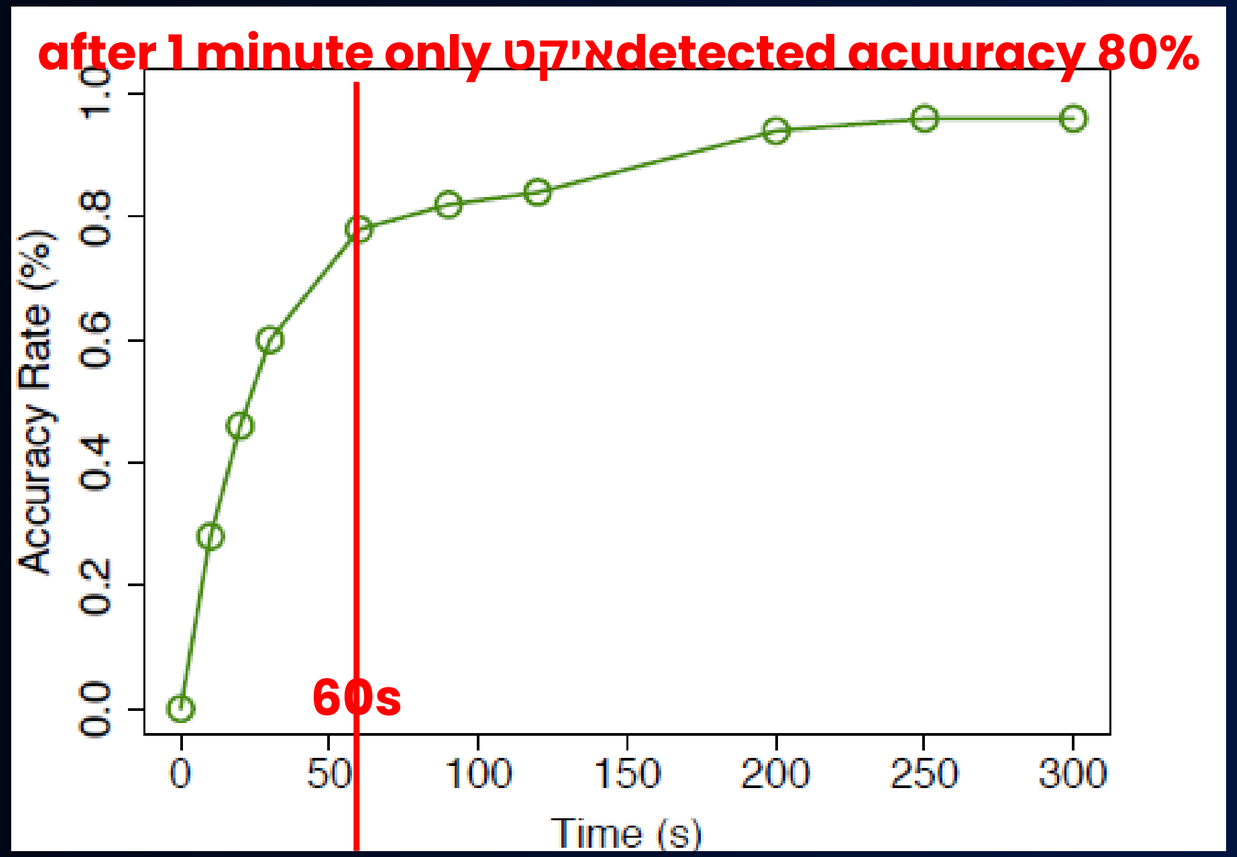
Detection accuracy quickly increases with time



RAPTOR Attack #1

Asymmetric routing

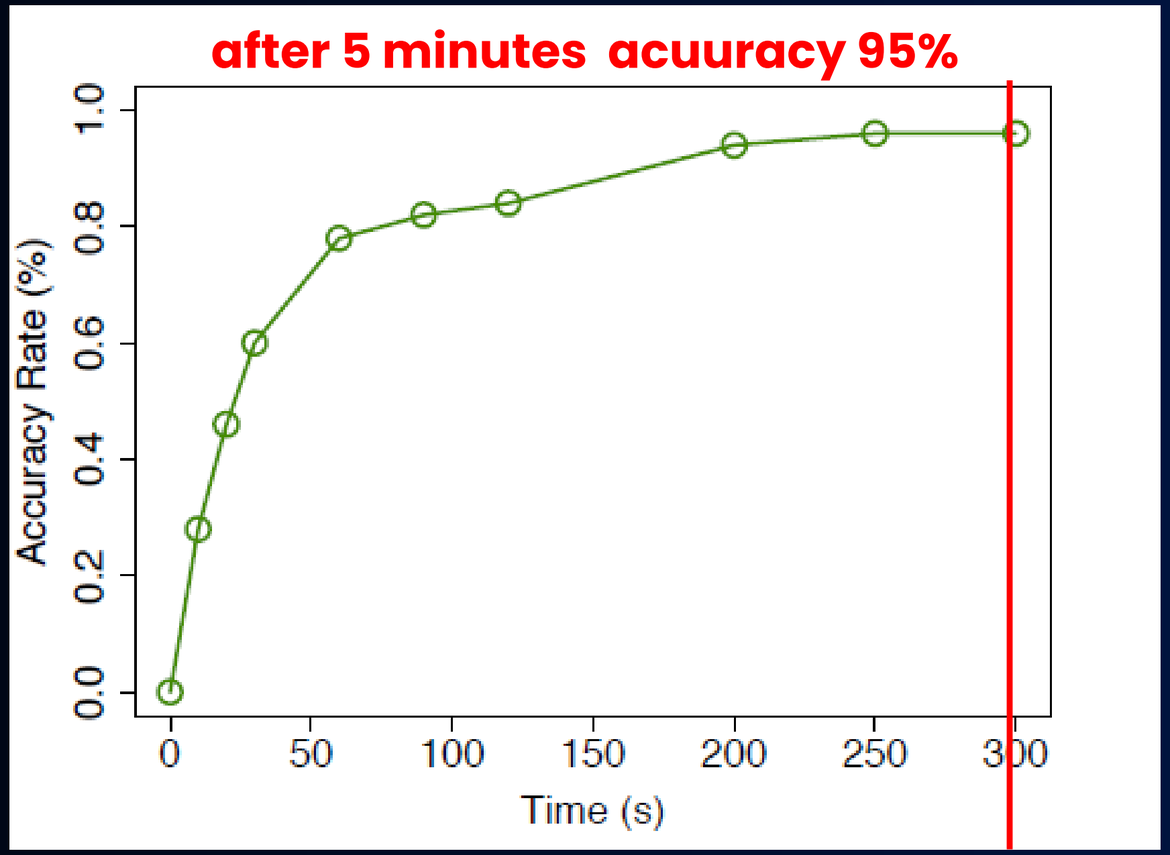
Detection accuracy quickly increases with time



RAPTOR Attack #1

Asymmetric routing

Detection accuracy quickly increases with time



- **Just ACKs from both ends were enough — as accurate as using full data packets**

RAPTOR Attack #2

BGP Churn

Experiment 2: BGP Churn and Growing AS Threat Over Time

RAPTOR Attack #2

BGP Churn

Phase 1 – Analysis of BGP Routing Data from January 1, 2015

they measured the effect of churn by collecting BGP updates for 1 month (Jan 15)

the initial dataset was then re-analyzed dynamically over the following month to observe changes and assess the impact on Tor circuit vulnerability

RAPTOR Attack #2

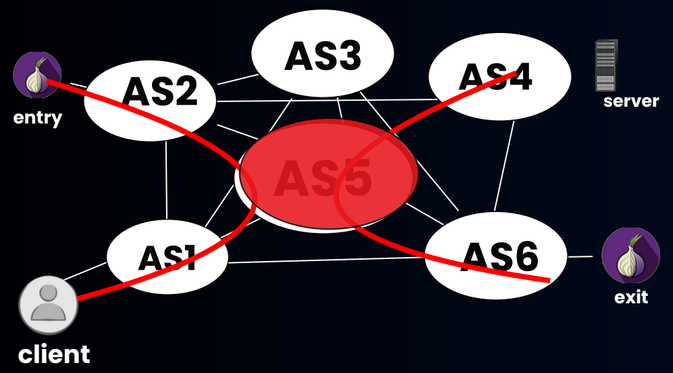
BGP Churn

What was analyzed - 1 janury

On the static baseline alone (Jan 1), 5,593 client–destination pairs were already compromisable

Phase 1 – Static Analysis

we look for ASes that behave like AS5 — appearing on both sides of the communication — and observe how many more such ASes emerge after one month of churn

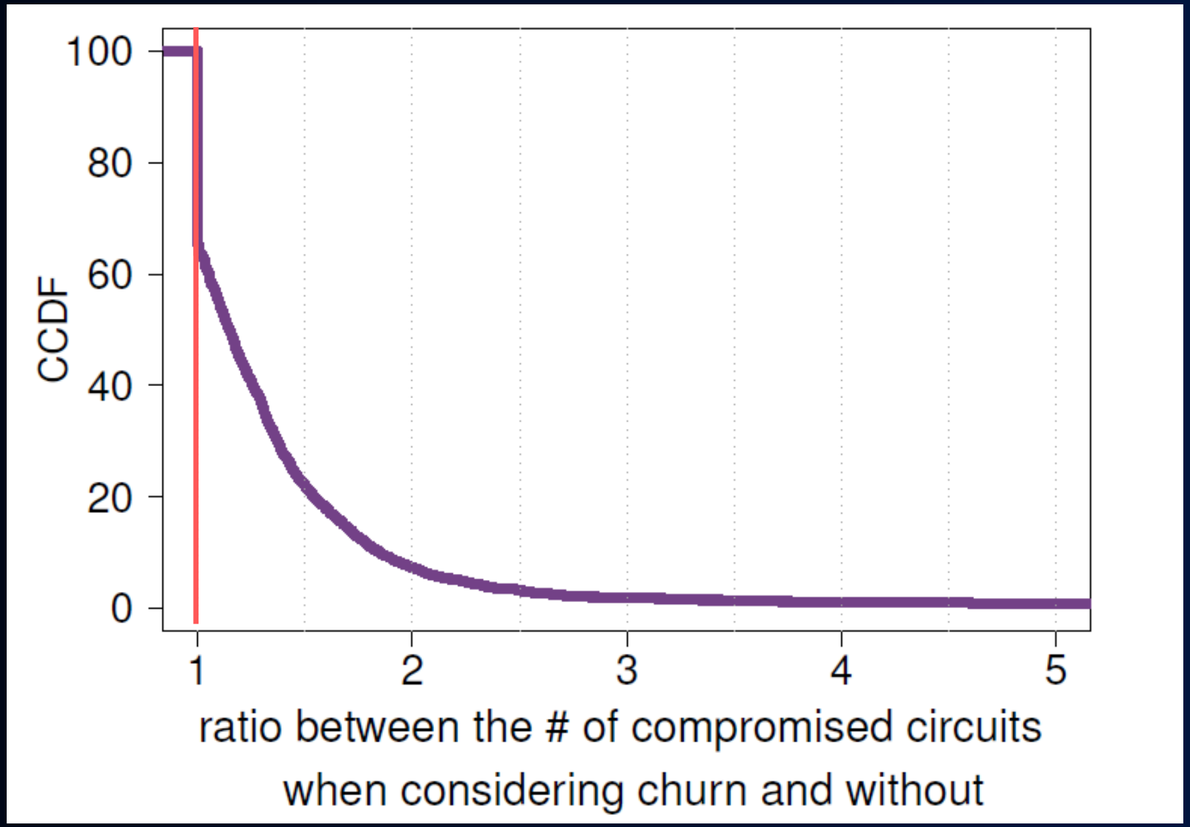


RAPTOR Attack #2

BGP Churn

Phase 2 — Dynamic Analysis of BGP Churn

After one month of BGP changes, 60% of the client–destination pairs had an increase in the number of vulnerable circuit

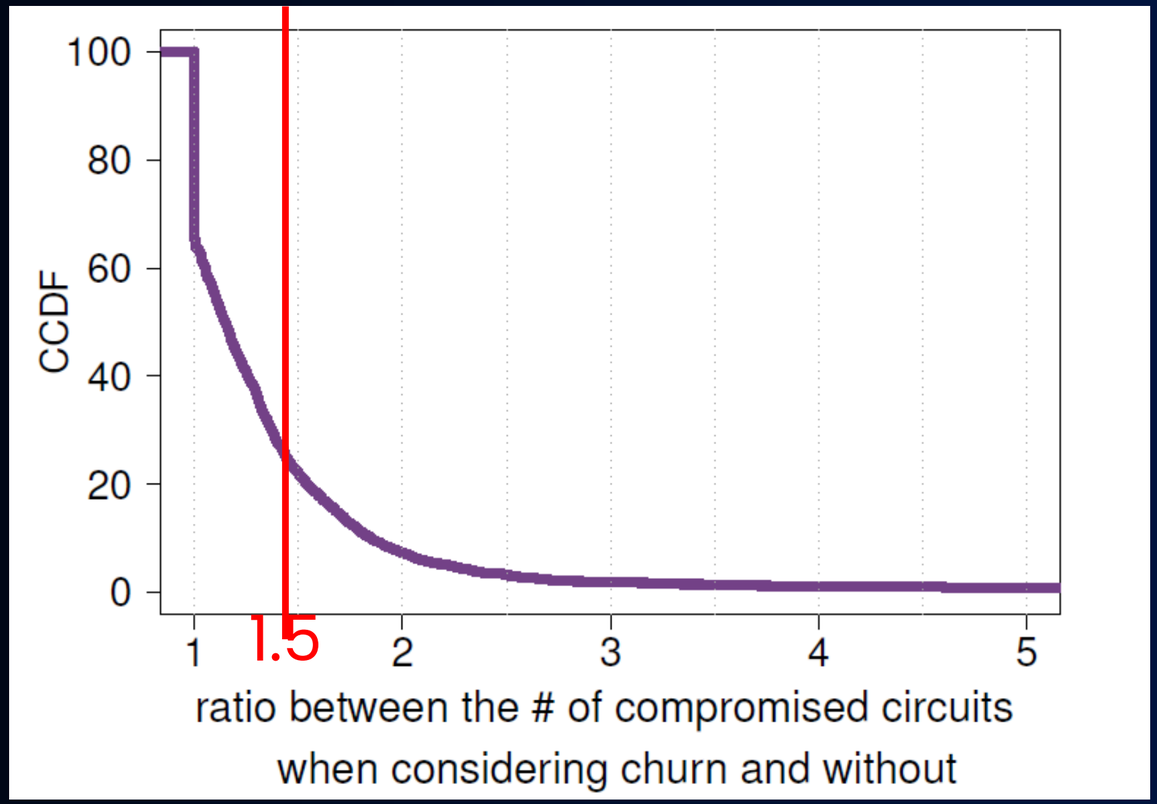


RAPTOR Attack #2

BGP Churn

Phase 2 — Dynamic Analysis of BGP Churn

20% of pairs see up to 50% increase of vulnerable circuits



- Total compromisable pairs rose from 5593 to 5754 (+3%)

RAPTOR Attack #2

BGP Churn

From Theory to Practice

The first experiment of bGP churn relied on routing tables to estimate the threat

Now we use live measurements to verify the real-world impact

+3% rise in compromisable pairs is just the beginning
– now we explore what that means in terms of circuit-level vulnerability across different directions of traffic

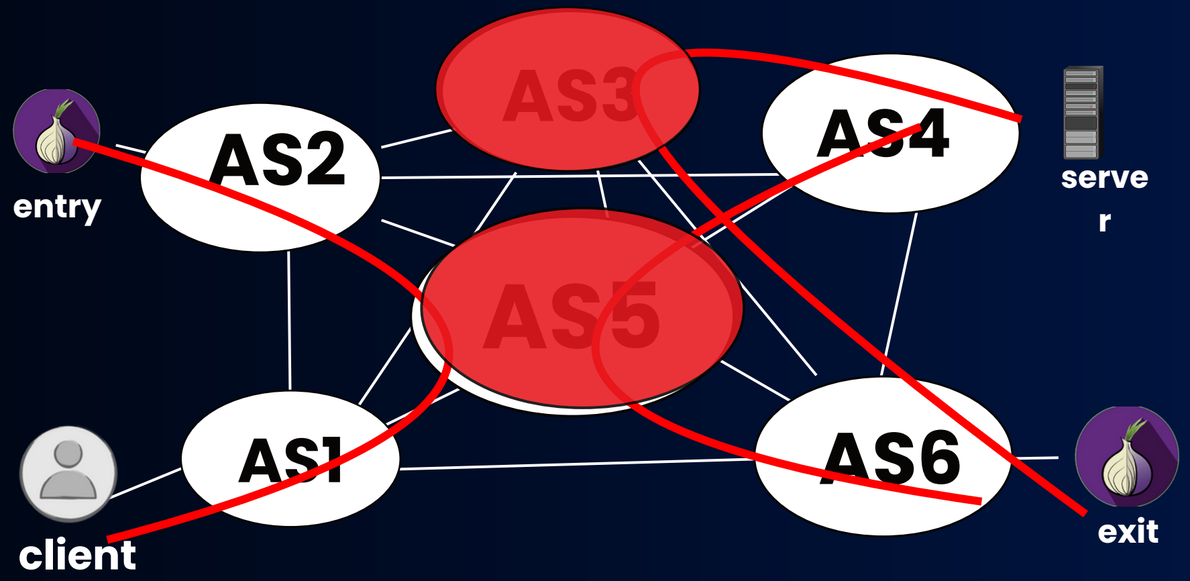
RAPTOR Attack #2

BGP Churn

Phase 3 – Live Measurements (Traceroute-based Verification)

AS-Level Threat Analysis

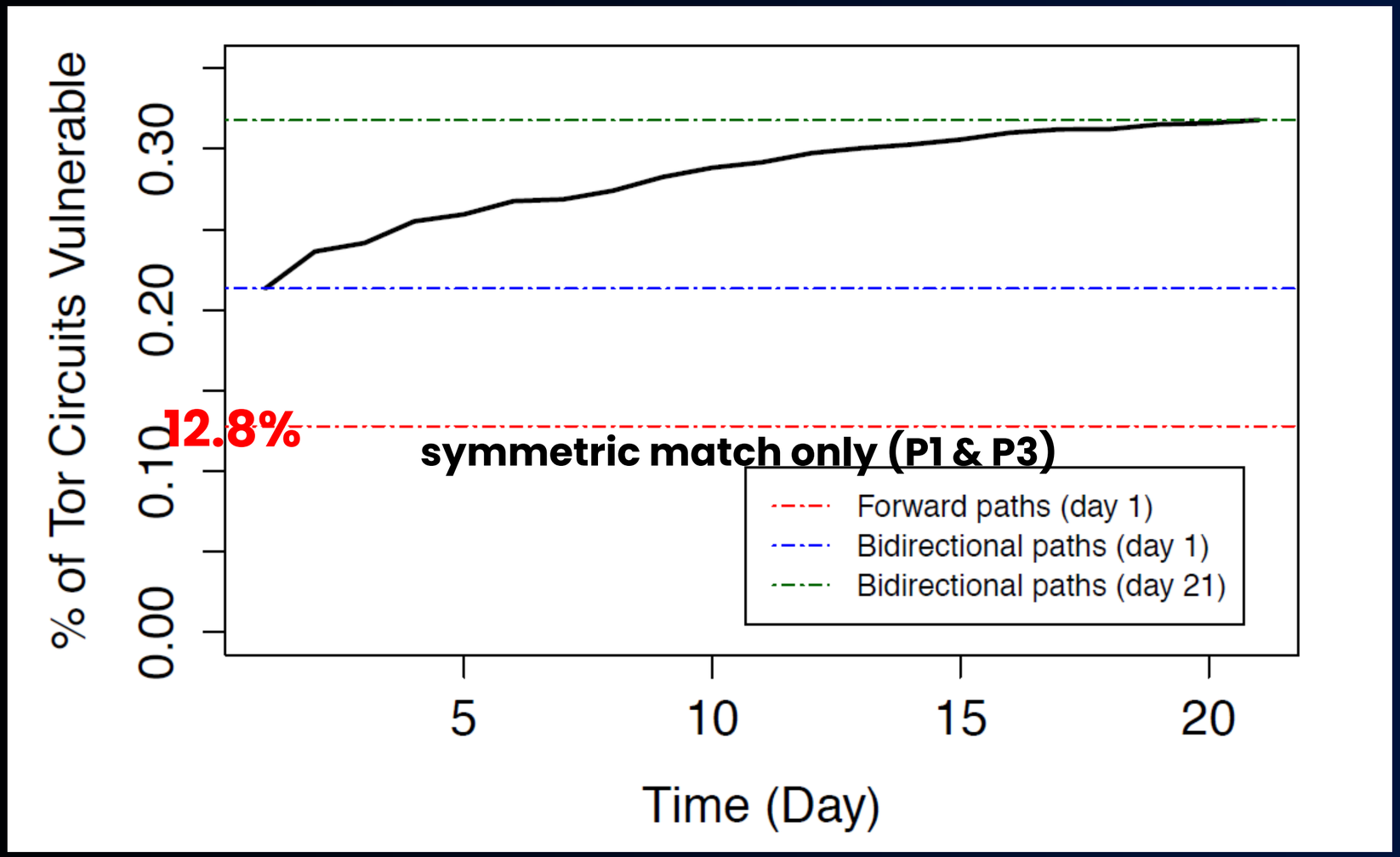
they identified all ASes that act like AS5 or AS3 — those that appear across multiple traffic directions and thereby compromise Tor circuits.



RAPTOR Attack #2

BGP Churn

Phase 3 – Live Measurements (Traceroute-based Verification)



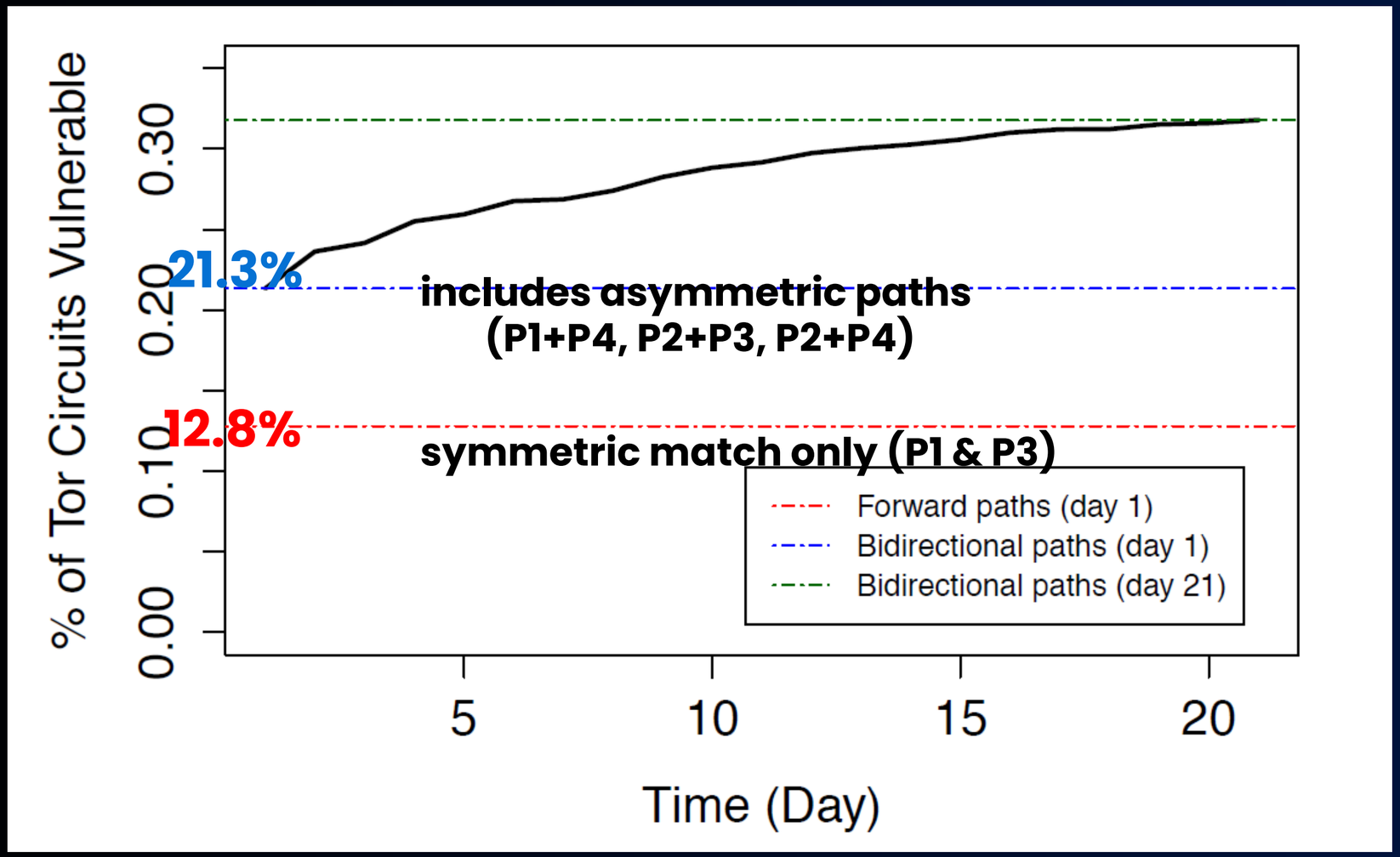
P1: Client → Guard

P3: Exit → Destination

RAPTOR Attack #2

BGP Churn

Phase 3 – Live Measurements (Traceroute-based Verification)



P2: Guard → Client
P4: Destination → Exit

P1: Client → Guard
P4: Destination → Exit

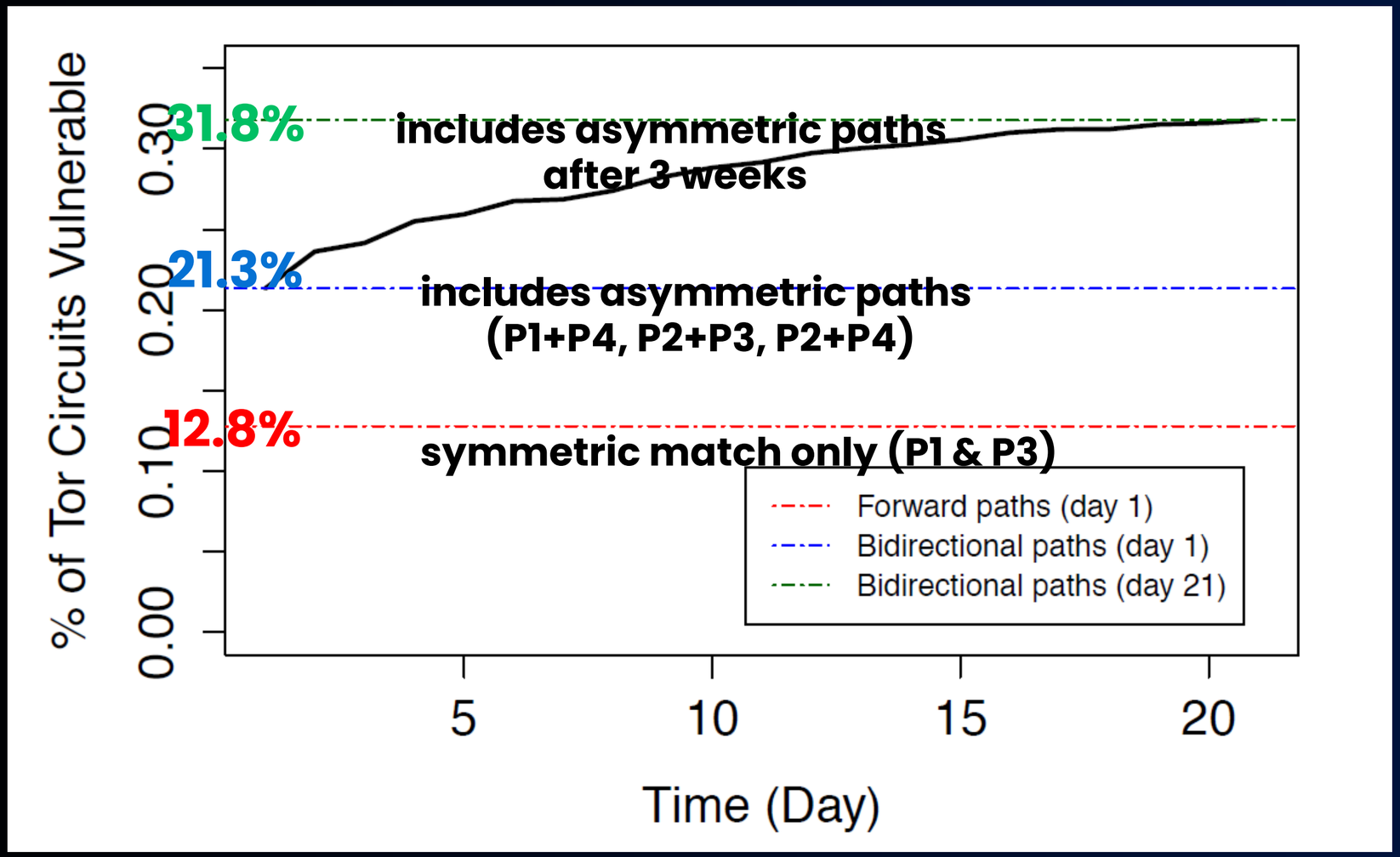
P2: Guard → Client
P3: Exit → Destination

P1: Client → Guard
P3: Exit → Destination

RAPTOR Attack #2

BGP Churn

Phase 3 – Live Measurements (Traceroute-based Verification)



P2: Guard → Client
P4: Destination → Exit

P1: Client → Guard
P4: Destination → Exit

P2: Guard → Client
P3: Exit → Destination

P1: Client → Guard
P3: Exit → Destination

RAPTOR Attack #2

BGP Churn

Phase 2 — Dynamic Analysis of BGP Churn

Not part of the core experiment but a major takeaway from the data

A small number of powerful ASes are present on up to 90% of all Tor circuits

Name	ASN	Tor circuits (%) seen	Country
NTT	2914	91	US
IIJ	2497	91	Japan
BroadbandONE	19151	91	US
Inet7	13030	91	CH
Level3	3356	88	US
Tinet	3257	86	DE
Cogent	174	63	US
Level3/GBLX	3549	58	US
TATA AMERICA	6453	53	US
TeliaSonera	1299	50	SWE

While Tor assumes traffic is distributed across many independent AS paths, the reality is different.

2014 – real hijacks

Event	# hijacked relays	# hijacked guards	# hijacked exits
Indosat 2011	5 (0.24%)	1 (0.15%)	4 (0.44%)
Indosat 2014	44 (0.80%)	38 (1.80%)	17 (1.65%)

Table 6: Summary statistics for known Indosat prefix hijacking events.

Putting the Attack into Practice

Experiment 1 – Asymmetric Traffic: What if an AS sees only one direction of the flow?
Experiment 2 – BGP Churn: Can routing changes over time expose more users to the same AS?

Experiment 3 –Real-World BGP Hijack Experiment

Unlike the previous passive observations,
this experiment demonstrates an active prefix hijack: an attacker temporarily reroutes real Tor traffic — and brings it back — to test how vulnerable the network is to real-time manipulation.

RAPTOR Attack #3

BGP Hijack

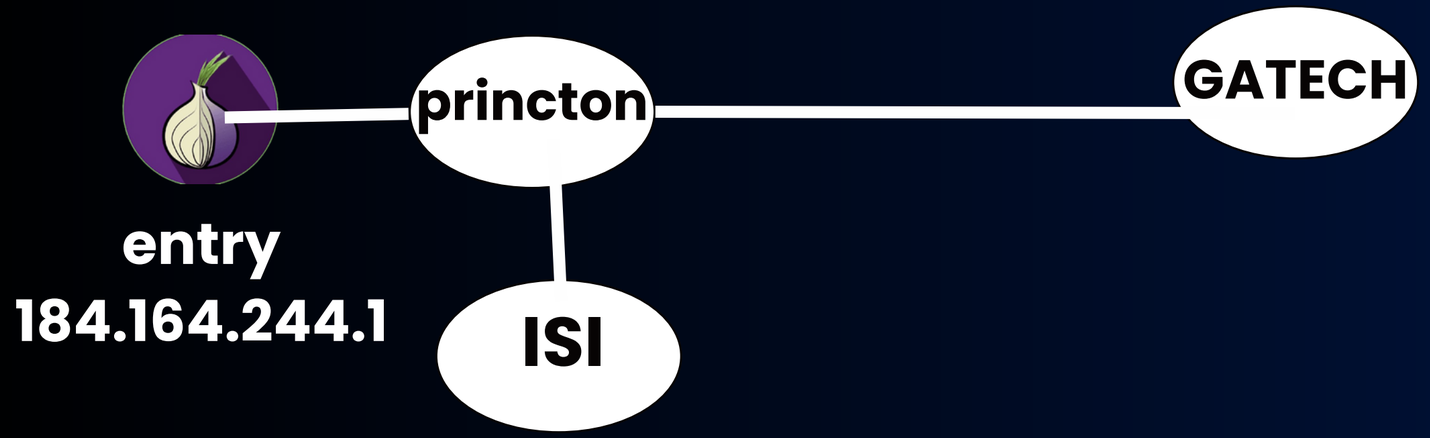
Can an Attacker Hijack a Tor Path?

RAPTOR Attack #3

BGP Hijack

Setup

- Tor entry relay hosted at Princeton.
- Announced via Georgia Tech AS using /23 prefix.

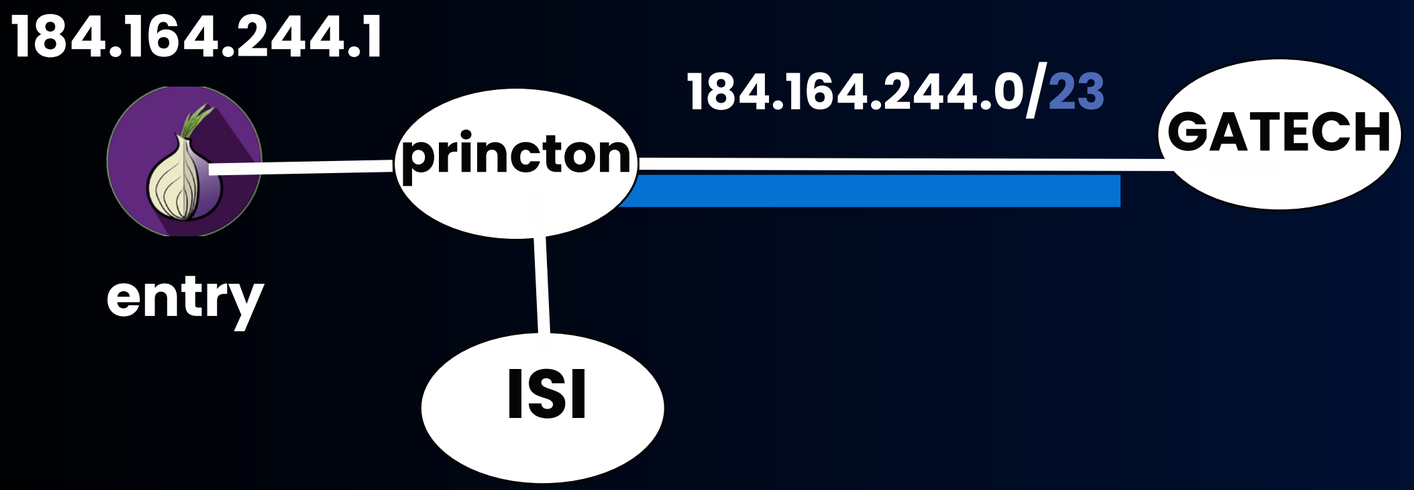


The relay was hosted at Princeton University using IP address 184.164.244.1, forming the target of the BGP hijack later in the experiment.

RAPTOR Attack #3

BGP Hijack

Selecting the Prefix for Hijack Testing



They announced the larger prefix 184.164.244.0/23 via Georgia Tech, enabling a later hijack using a more specific /24 prefix from another AS under their control.

RAPTOR Attack #3

BGP Hijack

BGP Prefix Advertisement via Georgia Tech

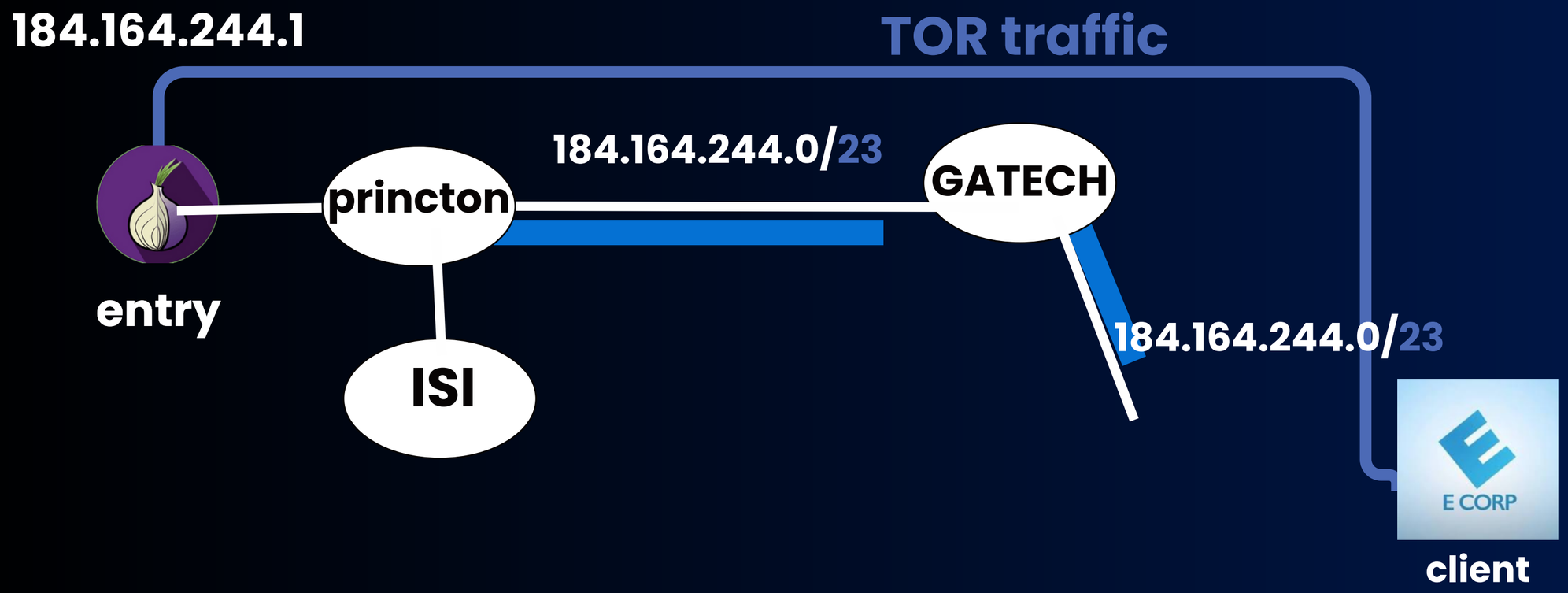


The /23 prefix was advertised from Georgia Tech and propagated across the Internet.
This made the Tor relay discoverable and routable by real clients, as if it were a normal relay.

RAPTOR Attack #3

BGP Hijack

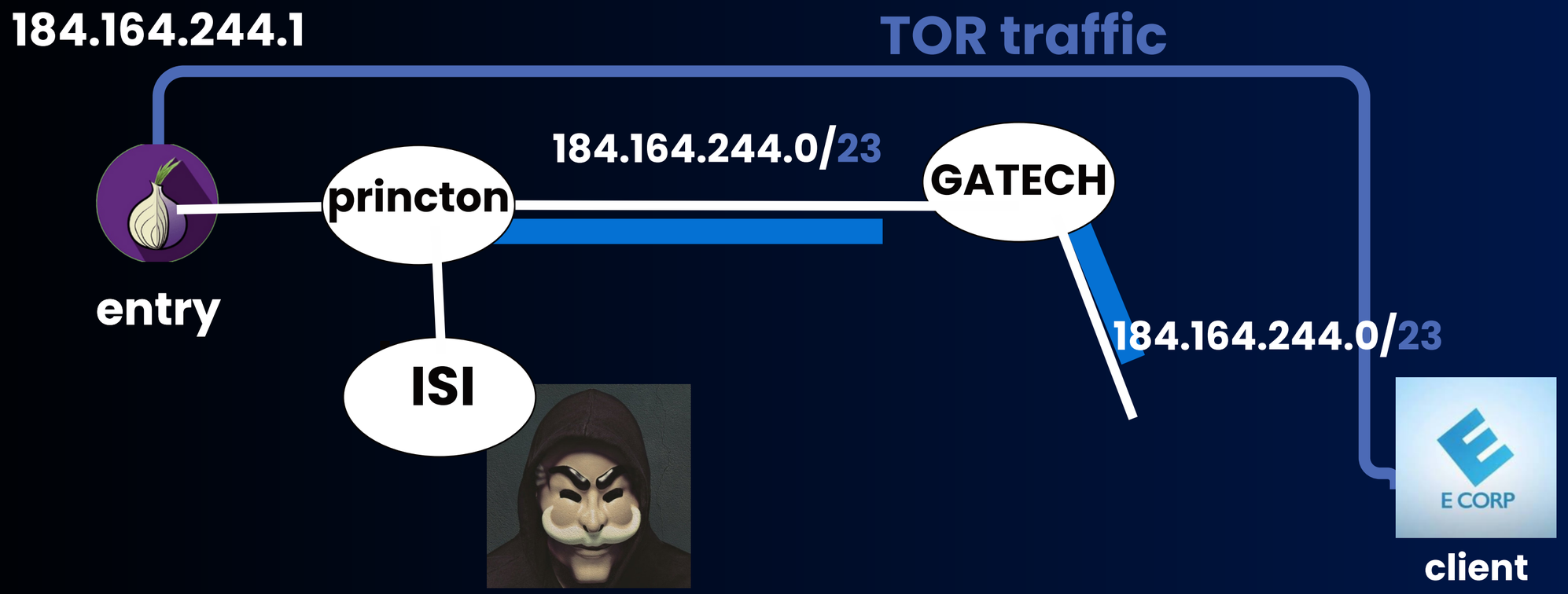
Real Tor Traffic Begins to Flow



After propagation, Tor clients began using the relay for real connections.
At this stage, the setup behaved exactly like any standard Tor entry node.

RAPTOR Attack #3

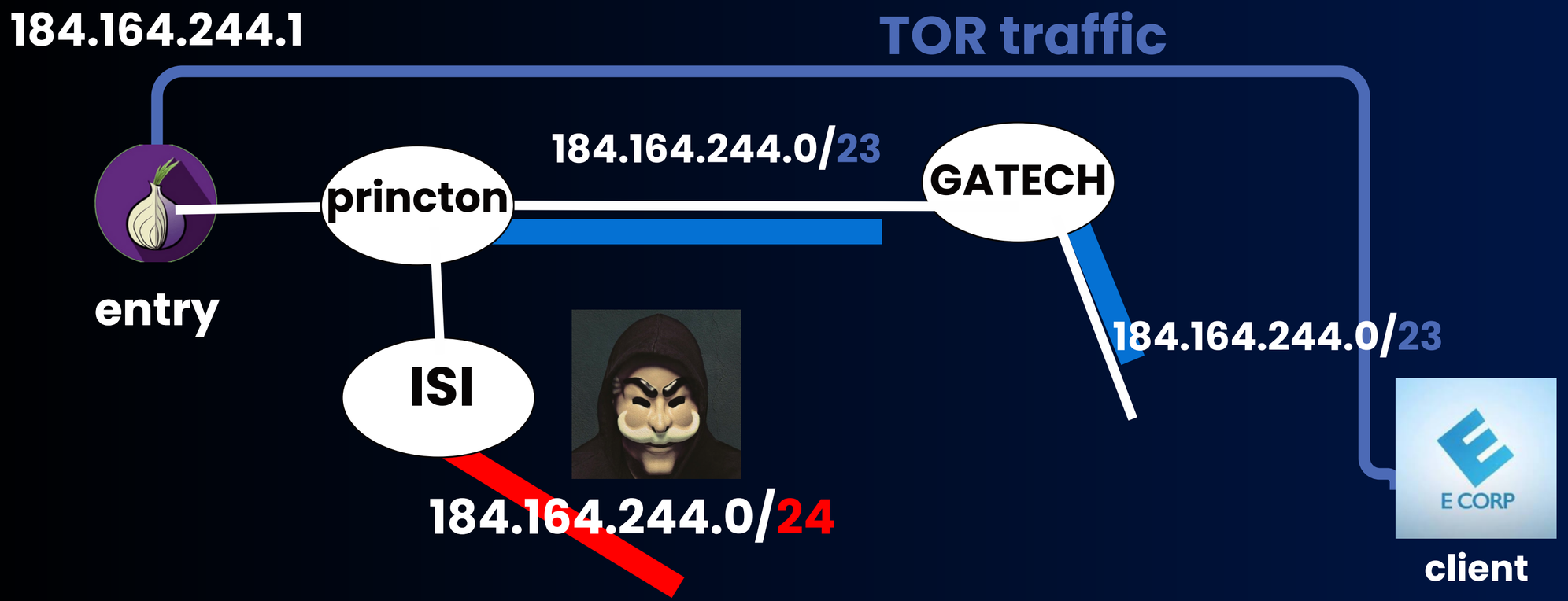
BGP Hijack



After ~20 seconds, they used a BGP speaker in ISI to announce the more-specific prefix

RAPTOR Attack #3

BGP Hijack

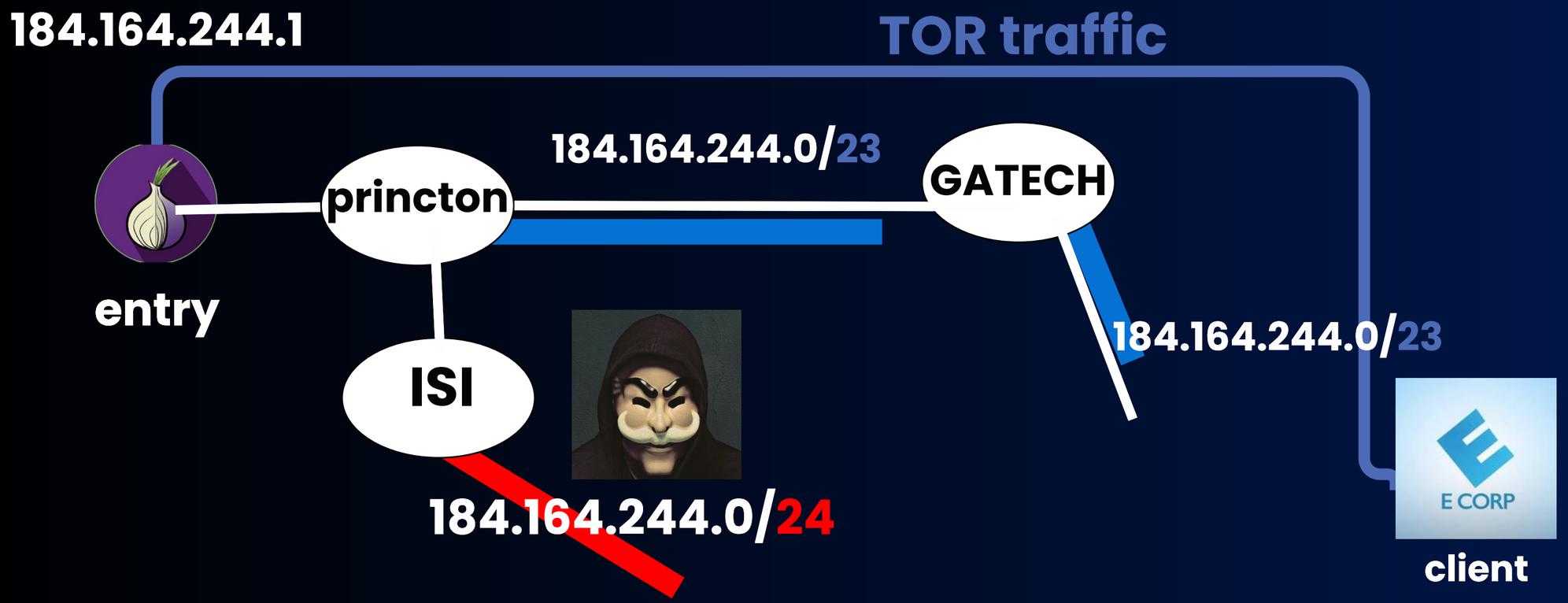


After ~20 seconds, they used a BGP speaker in ISI to announce the more-specific prefix

RAPTOR Attack #3

BGP Hijack

Traffic Rerouted via ISI

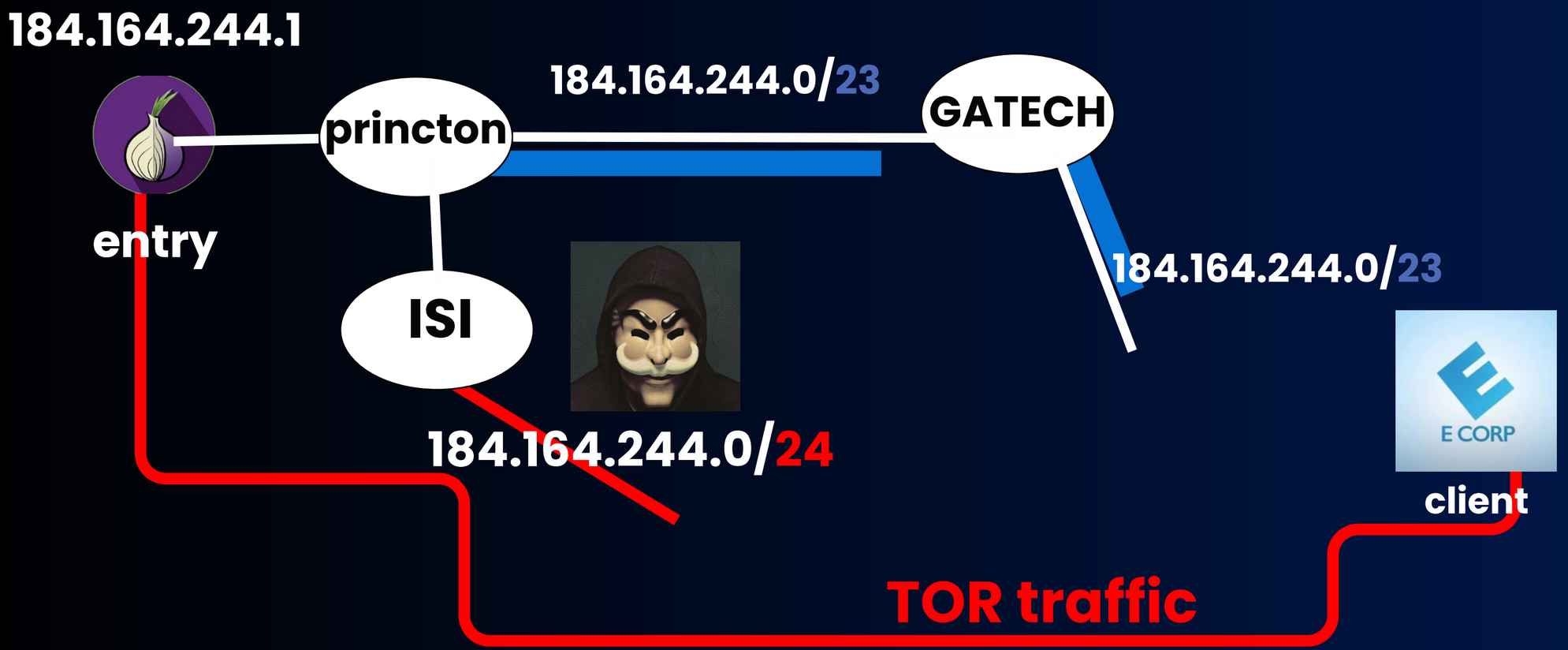


Due to BGP's Longest Prefix Match rule, traffic was now routed through ISI instead of Georgia Tech.

RAPTOR Attack #3

BGP Hijack

Traffic Rerouted via ISI

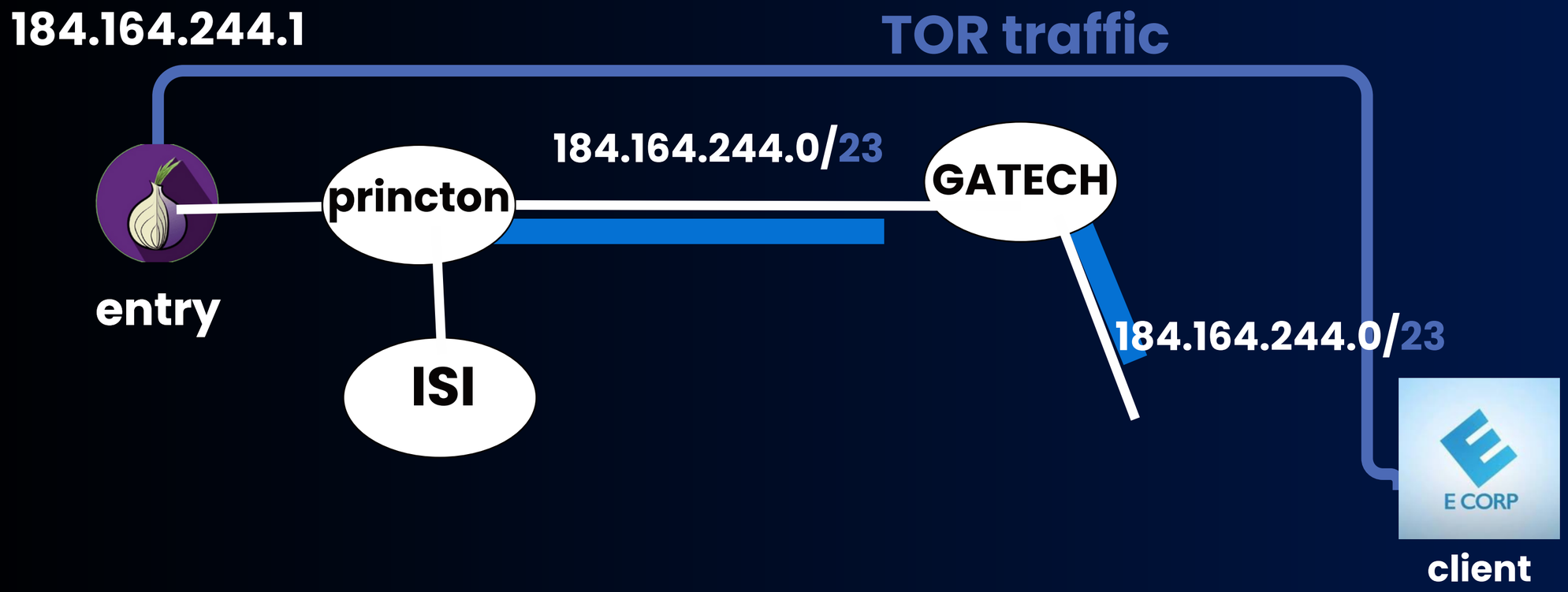


As forwarding is based on the longest-match,
all traffic soon started to enter via ISI

RAPTOR Attack #3

BGP Hijack

Hijack Ends After 300 Seconds



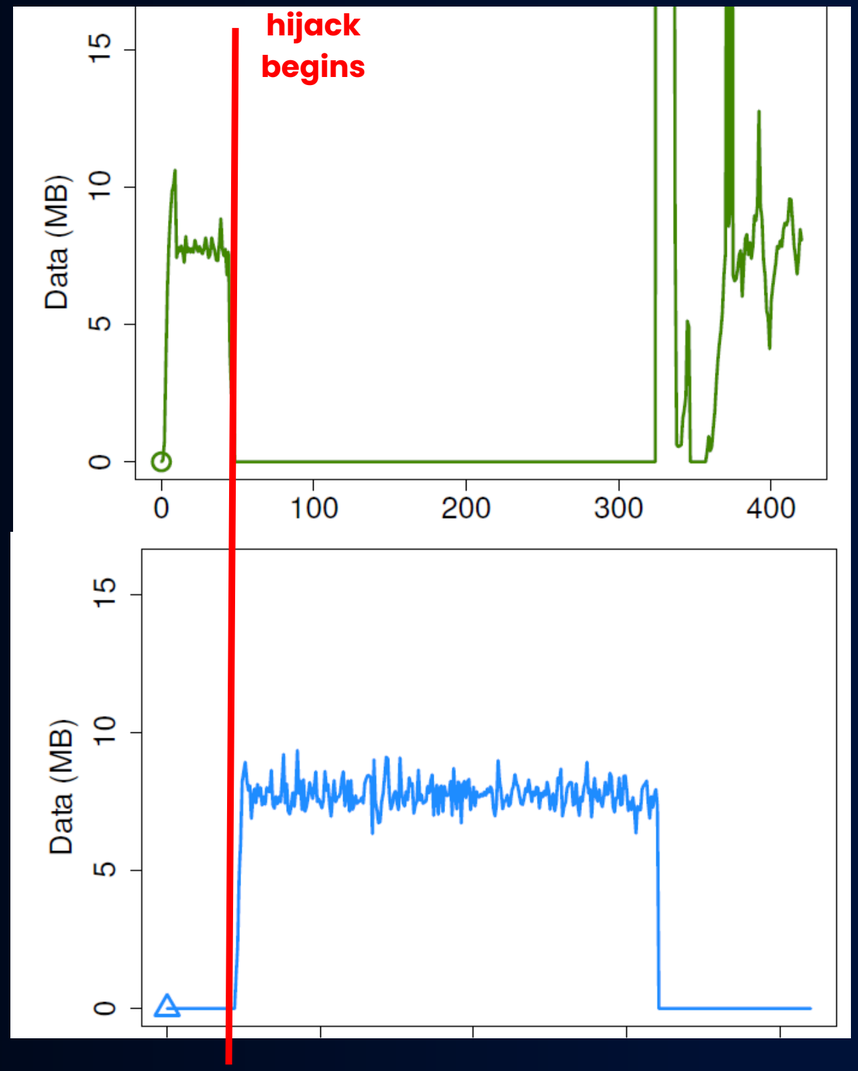
ISI Withdraws Announcement — Traffic Returns to Normal

RAPTOR Attack #3

BGP Hijack

Results of the Interception Attack

Traffic to entry



GATECH

Traffic to entry

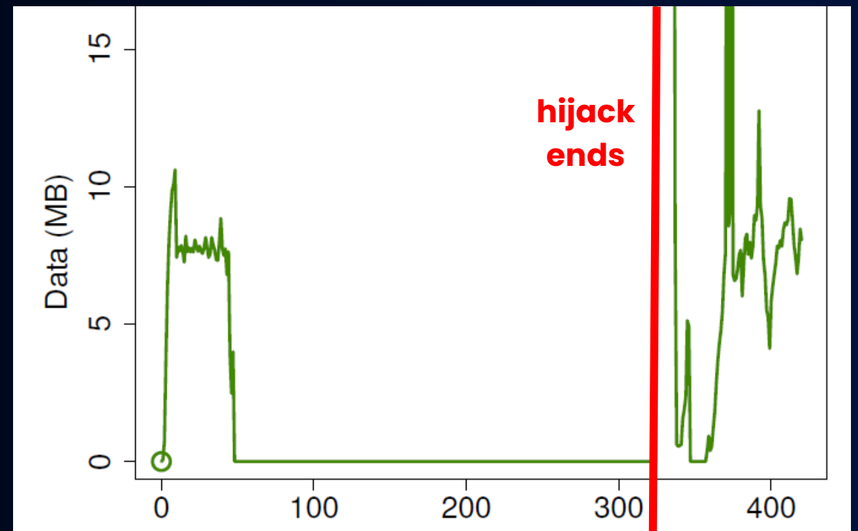
ISI

RAPTOR Attack #3

BGP Hijack

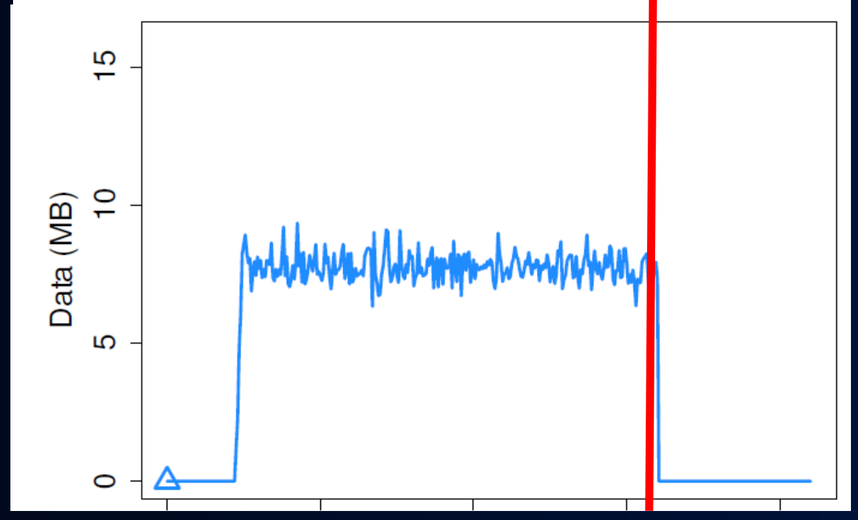
Results of the Interception Attack

Traffic to entry



GATECH

Traffic to entry



ISI

RAPTOR Attack #3

BGP Hijack

Verifying the Attack's Accuracy
(like in Experiment 1)

- The attacker (ISI) captures client ACKs after hijacking.
- Compare them to ACKs at the servers
- Same setup: 50 Tor clients → 50 web servers.

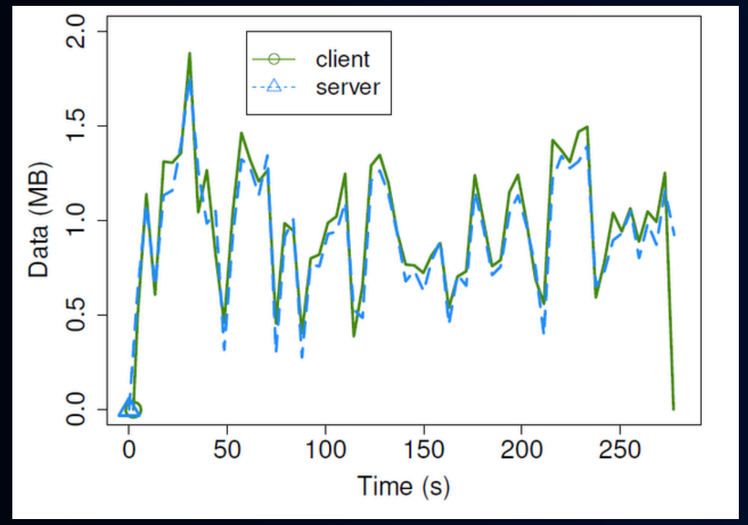
RAPTOR Attack #3

BGP Hijack

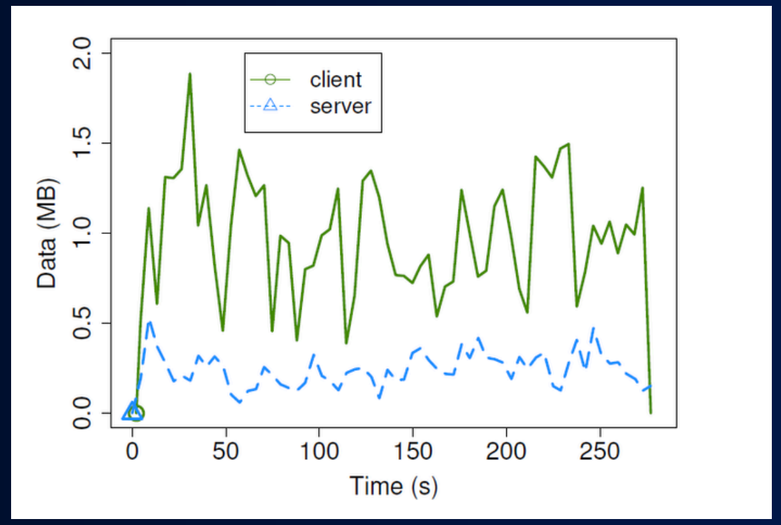
verifying the Attack's Accuracy

	Accuracy	False Negative	False Positive
Client ACK / Server ACK	90%	8%	2%

Successful



Failed Matches



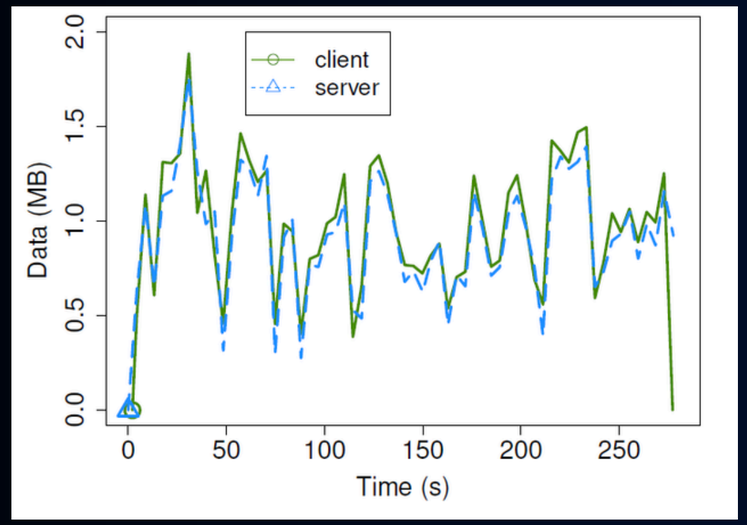
RAPTOR Attack #3

BGP Hijack

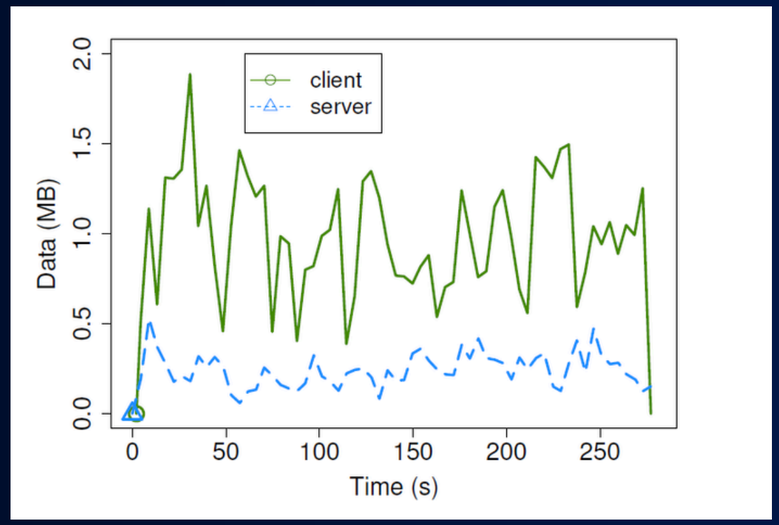
verifying the Attack's Accuracy

	Accuracy	False Negative	False Positive
Client ACK / Server ACK	90%	8%	2%

Successful



Failed Matches



RAPTOR Attack #3

BGP Hijack

Accuracy 90% lower than in Experiment 2 (95%).

Why?

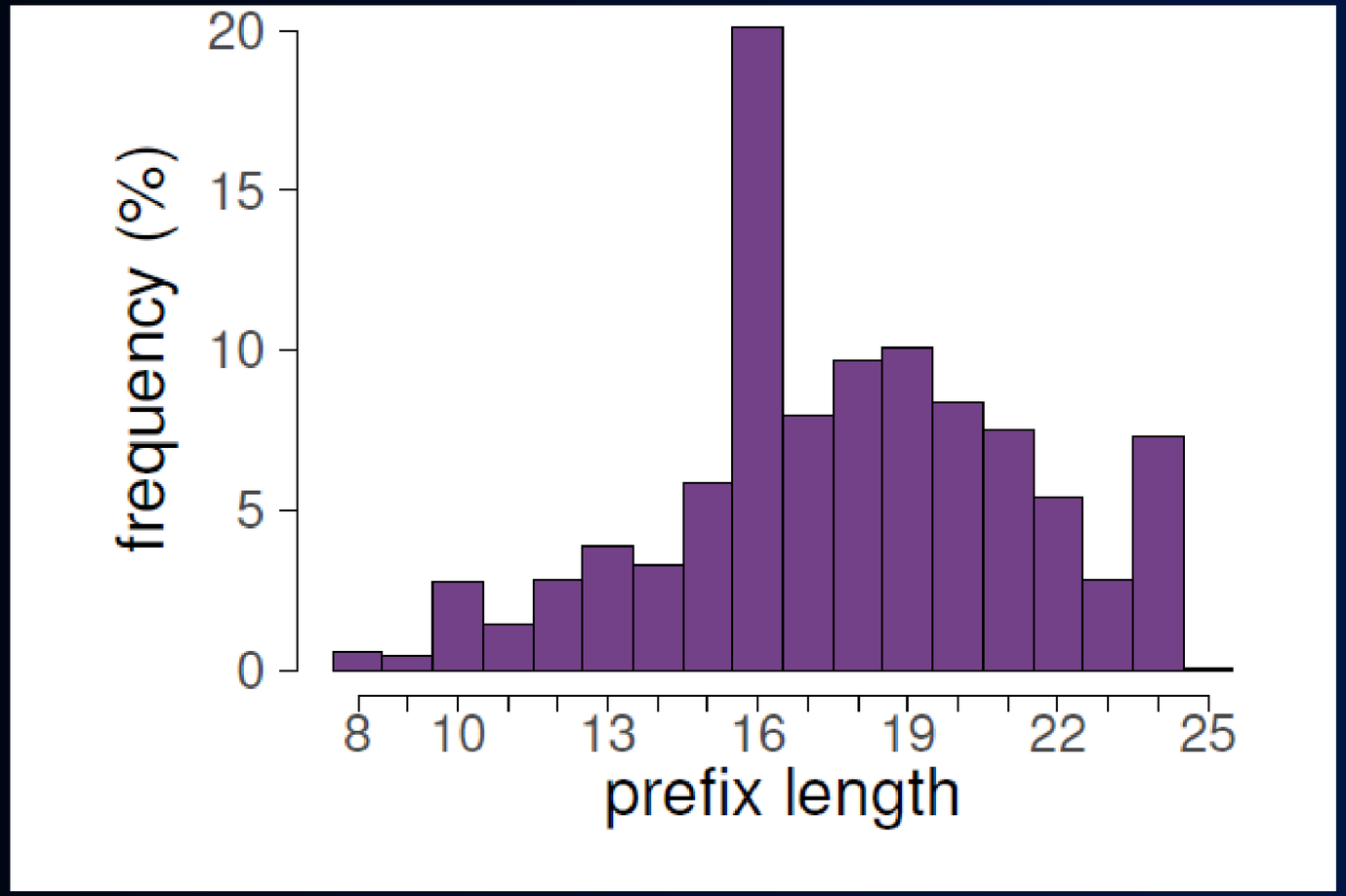
- All 50 clients used the same guard relay.
- Many also shared the same exit relay.
  - Similar network paths
  - Similar download sizes and timings
  - Synchronized ACK patterns
    - confusion between flows, increasing false negatives

This is an extreme case, unlikely in real Tor networks — it's important to show this because in the real world, **accuracy is typically 95% or higher**

RAPTOR Attack #3

BGP Hijack

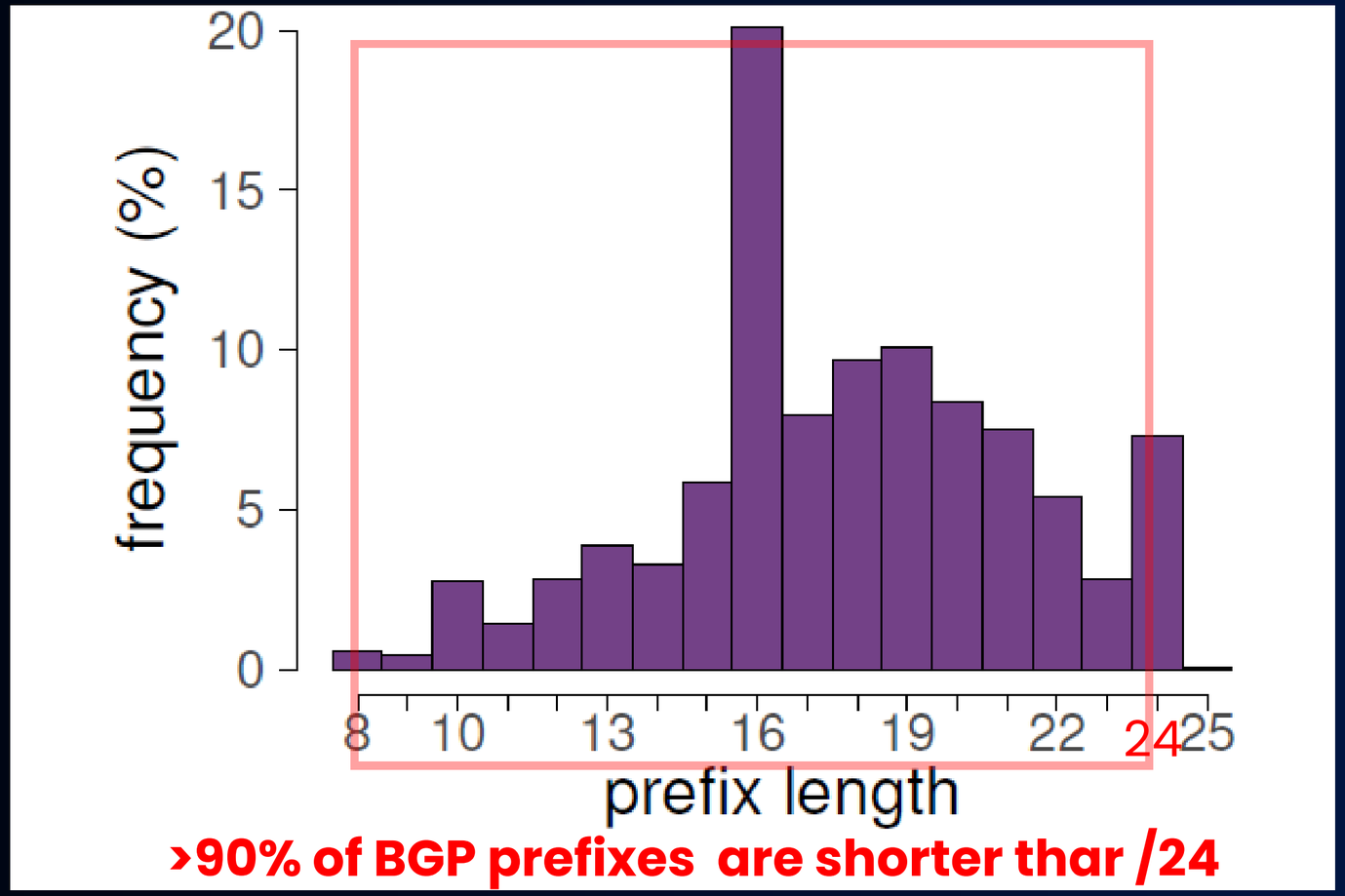
Tor Prefixes Are Vulnerable



RAPTOR Attack #3

BGP Hijack

Tor Prefixes Are Vulnerable



>90% of BGP prefixes are shorter thar /24

Quick Summary

Experiment 1 – Asymmetric Traffic: What if an AS sees only one direction of the flow?

Experiment 2 – BGP Churn: Can routing changes over time expose more users to the same AS?

Experiment 3 –Real-World BGP Hijack Experiment

Quick Summary

Experiment	Method	Findings
BGP Churn	Observed natural BGP updates over time	+3% more vulnerable circuits after 1 month
		60% of pairs saw increase
		20% had >50% more affected circuits
Asymmetric Traffic Analysis	50 clients & servers, captured TCP ACKs only	95% accuracy
		Using only one direction (ACKs)
BGP Prefix Hijack	Used Transit Portal to reroute Tor traffic via /24 prefix hijack	90% accuracy
		>90% of relays use hijackable prefixes (< /24);
		6 ASes host ~30% of guards/exits

countermeaseurs

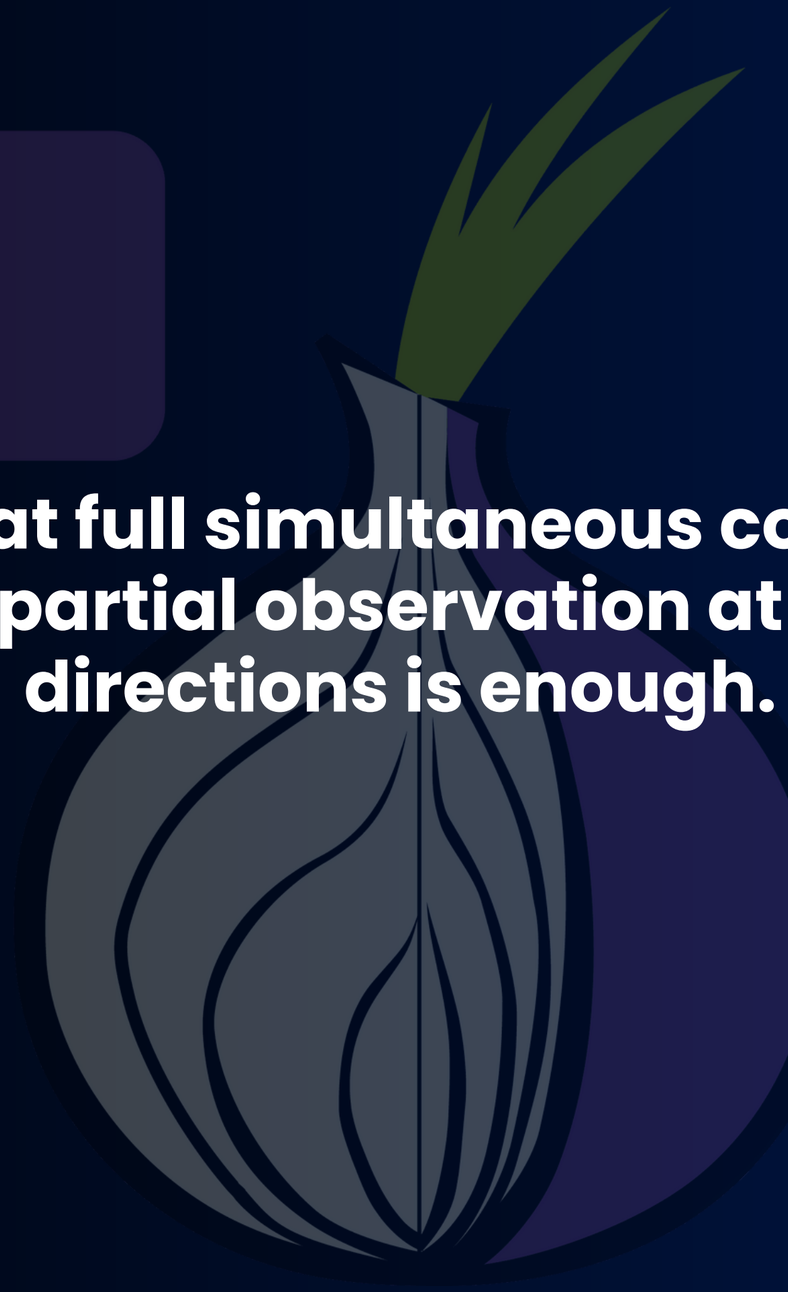
To protect itself, Tor should become more aware of the network underlying it

countermeaseurs

	countermeaseurs	Tools
Natural dynamism	look for stable relays	BGP monitoring
Route manipulation	Avoid suspicious relays, choose nearby ones	BGP monitoring and BGPsec
Asymmetric analsis	encrypt transport header	IPsec

countermeaseurs

	countermeaseurs	Tools	**disadventegs**
Natural dynamism	look for stable relays	BGP monitoring	**Less diversity → easier to fingerprint or analyze behavior**
Route manipulation	Avoid suspicious relays, choose nearby ones	BGP monitoring and BGPsec	**Can create predictable patterns → reduced anonymity**
Asymmetric analsis	encrypt transport header	IPsec	**Not widely used → may stand out, harder to deploy**



RAPTOR shows that full simultaneous control of both ends is not required — partial observation at different times or directions is enough.



QUESTIONS?